Exercises

Level - 1

(Practice Questions Based on Fundamentals)

ABC OF COMPLEX NUMBER

1. Find the value of $i^n + i^{n+1} + i^{n+2} + i^{n+3}$, $n \in I$.

2. Find the value of $i^{2010} + i^{2011} + i^{2012} + i^{2013}$.

3. Find the smallest integer n for which $\left(\frac{1+i}{1-i}\right)^n = 1$.

4. Find the sum of $\sum_{n=1}^{2013} (i^n + i^{n+1})$.

5. Find the value of $i^P + i^Q + i^R + i^S$ where P, Q, R, S are four consecutive integers.

6. Find the value of $i^{2015} + i^{2016} + i^{2017} + i^{2018}$.

7. If $\sum_{k=0}^{2016} i^k + \sum_{p=0}^{2018} i^P = x + iy, i = \sqrt{-1}$, find x + y + 2.

8. Find the smallest positive integer n for which $(1 + i)^{2n} = (1 - i)^{2n}$.

9. Find the value of $(1 + i)^5 + (1 + i^3)^5 + (1 + i^5)^7 + (1 + i^7)^7$.

10. Let $z = (n + i)^4$. Find the number of integral values of n for which z is an integer.

11. If $z = 1 + i$, find the multiplicative inverse of z^2.

12. If $z = \frac{1+2i}{3-4i}$, find the multiplicative inverse of z.

13. If $a + ib > c + id$, find the value of $b + d + 2016$.
14. Find the least positive integer n for which $\left(\frac{1+i}{1-i}\right)^n = \frac{2}{\pi}\left(\sin^{-1}x + \sec^{-1}\left(\frac{1}{x}\right)\right)$.
15. Find x and y which satisfy the equation $\frac{(1+i)x-2i}{(3+i)} + \frac{(2-3i)y+i}{(3-i)} = i$.
16. If $x + iy = \frac{2^{1008}}{(1+i)^{2016}} + \frac{(1+i)^{2016}}{2^{1008}}$, find x and y.
17. Let $z = x + iy$. If $z^{1/3} = a + ib$, prove that $\frac{x}{a} + \frac{y}{b} = 4(a^2 - b^2)$.
18. If $f(x) = x^4 - 8x^3 + 4x^2 + 4x + 39$ and $f(3 + 2i) = a + ib$, then find $\left(\frac{b}{a} + 10\right)$.
19. Find the least positive integer n for which $z = \left(\frac{2i}{1+i}\right)^n$ is a positive integer.
20. If $x = 3 + 2i$ is a root of a quadratic equation, find its equation.
21. Solve: $z^2 + \bar{z} = 0$.
22. If $(i - i)$ is the root of the equation $z^3 - 2(2 - i)z^2 + (4 - 5i)z + (3i - 1) = 0$, find the other roots.
23. Given that $1 + 2i$ is one root of the equation $x^4 - 3x^3 + 8x^2 - 7x + 5 = 0$, find the other three roots.

MODULUS OF COMPLEX NUMBERS

24. If $|z - (2 + 3i)| = 1$, find the greatest and the least value of $|z|$.
25. If $|z + 3 + 5i| = 2$, find the difference between the greatest and the least value of $|z|$.
26. If $a + ib = (1 + i)(1 + 2i)(1 + 3i) \ldots (1 + ni)$, then find the value of $a^2 + b^2$
27. If $x + iy = \frac{a+ib}{a-ib}$, prove that $x^2 + y^2 = 1$.
28. The complex number z satisfies $z + |z| = 2 + 8i$, find $|z|$.
29. If $z = re^{i\theta}$, then find $|e^{iz}|$.
30. If α, β be different complex numbers, find the maximum value of $\frac{\alpha\bar{\beta} + \beta\bar{\alpha}}{|\alpha\beta|}$.
31. If $|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1$, find the value of $|z_1 + z_2 + z_3|$.
32. If $z = (3 + 7i)(p + iq), p, q \in I - \{0\}$ is purely imaginary number, find the minimum value of $|z|^2$.
33. If α, β be different complex numbers with $|\beta| = 1$, find the value of $\left|\frac{\beta - \alpha}{1 - \bar{\alpha}\beta}\right|$.
34. Find the complex number z, if $|z + 1| = z + 2(1 + i)$.
35. If $|z - 1|^2 + |z + 1|^2 = 5$, find $|z|^2$.
36. If $|z - 2| = 2|z - 1|$, prove that $|z|^2 = \frac{4}{3}\text{Re}(z)$.
37. If $|z + 6| = |3z + 2|$, prove that $|z| = 2$.
38. Let $z = x + iy$ and $|z + 6| = |2z + 3|$, the locus of z is $x^2 + y^2 = 9$.
39. If $|z| = 1$ and $\omega = \frac{z-1}{z+1}$ (where $z \neq 1$), find $Re(\omega)$.
40. If z be a complex number satisfying the equation $|z + i| + |z - i| = 8$ on the complex plane, find the maximum value of $|z|$.

ARGUMENT OF COMPLEX NUMBERS

41. Find the arguments of
    (i) 1+i (ii) 1 – i
    (iii) –1 + i (iv) –1 – i
    (v) 0 (vi) 2013
    (vii) –2013 (viii) 2 i
    (ix) – 2 i (x) $\frac{1}{(1+i)}$
42. If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}$, find Arg($-z$).
43. If Arg(z) < 0, find the value of Arg(z) – Arg($-z$).
44. If $z = x + iy$ such that $|z + 1| = |z - 1|$ and $\text{Amp}\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$, find z.
45. If z_1 and z_2 are two non-zero complex numbers such that $|z_1 + z_2| = |z_1 - z_2|$, find $\text{Amp}\left(\frac{z_1}{z_2}\right)$.
46. Let $z = \left(\frac{\cos\theta + i\sin\theta}{\cos\theta - i\sin\theta}\right), \frac{\pi}{4} < \theta < \frac{\pi}{2}$, find Arg($z$).
47. Find the argument of z if $z = \sin\left(\frac{\pi}{5}\right) + i\left(1 - \cos\left(\frac{\pi}{5}\right)\right)$
48. If $\text{Arg}(z) = \frac{\pi}{3}$ and $\text{Arg}(z-1) = \frac{5\pi}{6}$, find the complex number z.
49. Let z be a complex number having the argument θ, $0 < \theta < \frac{\pi}{2}$ and satisfying the inequality $|z - 3i| = 3$, find $\text{Arg}\left(\cot\theta - \frac{6}{z}\right)$.
50. If $\text{Amp}\left(\frac{z-1}{z+1}\right) = \frac{\pi}{3}$, find the locus of z.
51. Find the angle that the vector representing the complex number $\frac{1}{(\sqrt{3}-i)^{25}}$ makes with the positive direction of the real axis.

MODULUS AND ARGUMENT OF COMPLEX NUMBERS

52. If $z_1, z_2 \in C$, prove that $|z_1 + z_2| \leq |z_1| + |z_2|$.
53. If $z_1, z_2 \in C$, prove that $||z_1| - |z_2|| \leq |z_1 - z_2|$.
54. If $|z_1 + z_2| = |z_1 - z_2| \Leftrightarrow \text{Arg}(z_1) - \text{Arg}(z_2) = \frac{\pi}{2}$.

55. If $|z_1 + z_2| = |z_1| + |z_2|$, prove that $\text{Arg}(z_1) = \text{Arg}(z_2)$.
56. If $|z_1 + iz_2| = |z_1| + |z_2| \Leftrightarrow \text{Arg}(z_1) - \text{Arg}(z_2) = \frac{\pi}{2}$.
57. If $|z_1 - z_2| = |z_1| + |z_2| \Leftrightarrow z_1 + kz_2 = 0, k \in I^+$
58. If $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \Leftrightarrow \frac{z_1}{z_2}$ is purely imaginary number.
59. If $|z_1| \le 1, |z_2| \le 1$, prove that
$|z_1 - z_2|^2 \le (|z_1| - |z_1|)^2 + (\text{Arg}(z_1) - \text{Arg}(z_2))^2$
60. If $|z_1| \le 1, |z_2| \le 1$, prove that
$|z_1 + z_2|^2 \le (|z_1| - |z_1|)^2 - (\text{Arg}(z_1) - \text{Arg}(z_2))^2$
61. Find the maximum value of $|z + 1|$, where $|z + 4| \le 3$.
62. Find the minimum value of $|z + 2|$, where $|z + 5| \le 4$.
63. Find the minimum values of
(i) $|z| + |z + 2|$
(ii) $|z + 2| + |z - 2|$
64. Find the maximum value of $|z + 2| + |z - 2| + |2z - 7|$.
65. Find the greatest and the least values of $|z_1 + z_2|$, where $z_1 = 6 + 8i$ and $z_2 = 3 + 4i$
66. If $|z - 3i| \le 4$, find the maximum value of $|i(z + 1) + 1|$.
67. If $|z| = 3$, then find the min and maximum values of $\left|z + \frac{1}{z}\right|$.
68. If $|z_1| = 2, |z_2| = 3, |z_3| = 5$ such that $|25z_1z_2 + 9z_1z_3 + 4z_2z_3| = 90$, find the value of $|z_1 + z_2 + z_3|$.

SQUARE ROOT OF A COMPLEX NUMBER

69. Find the square roots of $3 - 4i$.
70. Find the square roots of $5 + 12i$.
71. Find the square roots of $8 - 6i$.
72. Find the square roots of $3i$.
73. Find the square roots of $8 - 15i$.
74. Find the square roots of
$$x^2 + \frac{1}{x^2} + 4i\left(x - \frac{1}{x}\right) - 6.$$
75. If $z^2 + 5 = 12\sqrt{-1}$, find the complex number z.

CUBE ROOTS OF A COMPLEX NUMBER

76. If ω is the complex cube root of unity, find the value of $(2 + 3\omega + 3\omega^2)^{2013}$.
77. If ω is the complex cube root of unity, find the value of $(3 + 4\omega + 5\omega^2)^{10}$.
78. If ω is the non-real cube root of unity, find the sum of
$$\omega + \omega^{\left(\frac{1}{2}+\frac{3}{8}+\frac{9}{32}+\frac{27}{128}+\dots\right)}.$$
79. Find the common roots of $z^3 + 2z^2 + 2z + 1 = 0$ and $z^{2013} + z^{2014} + z^{2015} = 0$.
80. If α, β, γ be the cube roots of (-2013), for any x, y, and z, find the value of $\dfrac{x\alpha + y\beta + z\gamma}{x\beta + y\gamma + z\alpha}$.
81. Find the value of $\dfrac{2 + 3\omega + 4\omega^2}{4 + 3\omega^2 + 2\omega}$.
82. Find the value of $\dfrac{5 + 6\omega + 7\omega^2}{7 + 6\omega^2 + 5\omega} + \dfrac{5 + 6\omega + 7\omega^2}{6 + 5\omega + 7\omega^2}$.
83. If $i = \sqrt{-1}$, find the value of
$$4 + 5\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{365}.$$
84. Solve for x: $x^6 - 9x^3 + 8 = 0$.

DE MOIVRES THEOREM

85. If x satisfies the equation $x^2 - 2x\cos\theta + 1 = 0$, find the value of $x^n + \dfrac{1}{x^n}$.
86. If $z_r = \cos\left(\frac{2r\pi}{5}\right) + i\sin\left(\frac{2r\pi}{5}\right)$, $r = 1, 2, 3, 4, 5$, find the value of $z_1z_2z_3z_4z_5$.
87. If $\sin\alpha + \sin\beta + \sin\gamma = 0 = \cos\alpha + \cos\beta + \cos\gamma$, find the value of
(i) $\sin 3\alpha + \sin 3\beta + \sin 3\gamma$
(ii) $\cos 3\alpha + \cos 3\beta + \cos 3\gamma$
88. If $\sin\alpha + \sin\beta + \sin\gamma = 0 = \cos\alpha + \cos\beta + \cos\gamma$, find the value of
(i) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma$ (ii) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$
89. If $z_r = \cos\left(\frac{\pi}{2^r}\right) + i\sin\left(\frac{\pi}{2^r}\right)$, find the value of $z_1 \cdot z_2 \cdot z_3 \dots$ to ∞.
90. Find the value of $(1 + i)^8 + (1 - i)^8$.
91. Find the value of
$$\sum_{k=1}^{21}\left(\sin\left(\frac{2\pi k}{11}\right) - i\cos\left(\frac{2\pi k}{11}\right)\right).$$

*n*TH ROOTS OF COMPLEX NUMBERS

92. Solve for x: $x^3 - 1 = 0$.
93. Solve for x: $x^5 - 1 = 0$.
94. Solve for x: $x^7 - 1 = 0$.
95. Solve for x: $x^3 + 1 = 0$.
96. Solve for x: $x^5 + 1 = 0$.
97. Solve for x: $x^7 + 1 = 0$.
98. Solve for x: $x^{10} - 1 = 0$.
99. Solve for x: $x^{10} + x^5 + 1 = 0$.
100. Solve for x: $x^{10} - x^9 + x^8 - x^7 + \dots + x^2 - x + 1 = 0$.
101. Solve for z : $z^5 + 1 = 0$ and deduce that
$$4\sin\left(\frac{\pi}{10}\right)\cos\left(\frac{\pi}{5}\right) = 1.$$
102. Solve for z: $z^7 - 1 = 0$ and deduce that
$$\cos\left(\frac{\pi}{7}\right)\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right) = \frac{1}{8}.$$
103. Find the integral solutions of $(1 - i)^x = 2^x$.
104. If $z = \left(\frac{\sqrt{3} - i}{2}\right)$, find the value of $(z^{101} + z^{103})^{106}$.
105. Let $z = x + iy$ be a complex number, where x and y are integers. Find the area of the rectangle whose vertices are the roots of $\bar{z}z^3 + z\bar{z}^3 = 350$.

106. Let $z = \cos\theta + i\sin\theta$, find the value of
$\sum_{m=1}^{15} \mathrm{Im}(z^{2m-1})$ at $\theta = 2°$.

107. Let a complex number α, $\alpha \neq 1$ be a root of an equation $z^{p+q} - z^p - z^q + 1 = 0$, where p and q are distinct primes. Show that either
$1 + \alpha + \alpha^2 + \ldots + \alpha^{p-1} = 0$ or
$1 + \alpha + \alpha^2 + \ldots + \alpha^{q-1} = 0$ but not both.

ROATATION

108. If a point $P(3, 4)$ is rotated through an angle of 90° in anti-clockwise sense about the origin, find the new position of P.

109. If a point $Q(3, 4)$ is rotated through an angle of 180° in anti-clockwise sense about the origin, find the new position of Q.

110. If a point $P(3, 4)$ is rotated through an angle of 30° in anti-clockwise sense about the point $Q(1, 0)$, find the new position of P.

111. The complex number $\sqrt{3} + i$ becomes $-1 + i\sqrt{3}$ after rotating an angle θ about the origin in anti-clock-wise sense, find the angle θ.

112. The three vertices of a triangle are represented by the complex numbers 0, z_1, z_2. If the triangle is equilateral triangle, prove that $z_1^2 + z_2^2 = z_1 z_2$

113. If the origin and the roots of $z^2 + az + b = 0$ form an equilateral triangle, prove that $a^2 = 3b$.

114. If the area of a triangle on the complex plane formed by the complex numbers z, iz and $z + iz$ is 50 sq.u., find $|z|$.

115. If the area of a triangle on the complex plane formed by the complex numbers z, ωz, $z + \omega z$ is $16\sqrt{3}$ sq.u., find the value of $(|z|^2 + |z| + 2)$.

116. Let z_1 and z_2 be the nth roots of unity which subtend a right angle at the origin, prove that $n = 4k$, where $k \in N$.

117. Suppose z_1, z_2, z_3 be the vertices of an equilateral triangle inscribed in the circle $|z| = 2$. If $z_1 = 1 + i\sqrt{3}$, prove that $z_2 = -2$ and $z_3 = 1 - i\sqrt{3}$.

118. If a and b be real numbers between 0 and 1 such that the points $z_1 = a + i$, $z_2 = 1 + ib$ and $z_3 = 0$ form an equilateral triangle, prove that $a = 2 - \sqrt{3} = b$.

119. The adjacent vertices of a regular polygon of n-sides are the points z and its conjugate $\bar{z}$.
If $\dfrac{\mathrm{Im}(z)}{\mathrm{Re}(z)} = \sqrt{2} - 1$, find the value of n.

120. The vertices A and C of a square are $ABCD$ are $2 + 3i$ and $3 - 2i$ respectively. Find the vertices B and D.

121. A, B, C are the vertices of an equilateral triangle whose centre is i. If A represents the complex number $-i$, find the vertices B and C.

122. A man walks a distance of 3 units from the origin towards the north-east (N45°E) direction. From there, he walks a distance of 4 units towards the north-west (N45°W) direction to reach a point P. Find the position of the point P in the Argand plane.

123. A particle P starts from the point $z_0 = 1 + 2i$, where $i = \sqrt{-1}$. It moves first horizontally away from origin by 5 units and then vertically away from the origin 3 units to reach a point z_1. From z_1 the particle moves $\sqrt{2}$ units in the direction of the vector $\hat{i} + \hat{j}$ and then through an angle $\dfrac{\pi}{2}$ in anti-clockwise direction on a circle with the centre at the origin to reach a point z_2. Find the point z_2.

124. Let z_1 and z_2 be the roots of the equation $z^2 + pz + q = 0$, where the co-efficients p and q may be complex numbers. Let A and B be represent z_1 and z_2 in the complex plane. If $\angle AOB = \alpha$, and $OA = OB$, where O is the origin, prove that $p^2 = 4q\cos^2\left(\dfrac{\alpha}{2}\right)$.

LOCI OF A COMPLEX NUMBER

125. Find the locus of z, if $\mathrm{Arg}\left(\dfrac{z-1}{z+1}\right) = \dfrac{\pi}{4}$.

126. Find the locus of z, if $|z - 1| + |z + 1| \leq 4$.

127. Find the locus of z, if $|z - 2| + |z + 2| \leq 4$.

128. Find the locus of z, if $z = t + 5 + i\sqrt{4 - t^2}$, $t \in R$.

129. If $\left(\dfrac{z^2}{z-1}\right)$ is always real, find the locus of z.

130. If $\mathrm{Re}\left(\dfrac{1}{z}\right) = c, c \neq 0$, find the locus of z.

131. If $|z^2 - 1| = |z|^2 + 1$, find the locus of z.

Hints and Solutions

Level I

1. We have,

$i^n + i^{n+1} + i^{n+2} + i^{n+3}$
$= i^n(1 + i + i^2 + i^3)$
$= 0$

2. We have,

$i^{2010} + i^{2011} + i^{2012} + i^{2013}$
$= i^{2010}(1 + i + i^2 + i^3)$
$= 0$

3. We have,

$$\left(\frac{1+i}{1-i}\right)^n = 1$$

$$\Rightarrow \left(\frac{1+i}{1-i} \times \frac{1+i}{1+i}\right)^n = 1$$

$$\Rightarrow \left(\frac{(1+i)^2}{1^2-(-i)^2}\right)^n = 1$$

$$\Rightarrow \left(\frac{1+i^2+2i}{1+1}\right)^n = 1$$

$\Rightarrow (i)^n = 1$

$\Rightarrow n = \ldots, -8, -4, 0, 4, 8, 12, \ldots$

Clearly, n is not defined

> ***Notes***
> 1. The smallest positive integer n for which $\left(\frac{1+i}{1-i}\right)^n = 1$, then $n = 4$
> 2. The smallest non negative integer n for which $\left(\frac{1+i}{1-i}\right)^n = 1$, then $n = 0$
> 3. The smallest positive integer n for which $\left(\frac{1+i}{1-i}\right)^n$ is real, then $n = 2$.

4. We have,

$$\sum_{n=1}^{2013} (i^n + i^{n+1})$$

$$= \left(\underbrace{i + i^2 + i^3 + i^4 + \ldots + i^{2012}}_{=0} + i^{2013}\right) + \left(\underbrace{i^2 + i^3 + i^4 + i^5 + \ldots + i^{2013}}_{=0} + i^{2014}\right)$$

$= i^{2013} + i^{2014}$
$= i + i^2$
$= i - 1$

5. We have,

$i^P + i^Q + i^R + i^S$
$= i^P(1 + i + i^2 + i^3)$
$= i^P \times 0$
$= 0$

6. We have,

$i^{2015} + i^{2016} + i^{2017} + i^{2018}$
$= i^{2015}(1 + i + i^2 + i^3)$
$= i^{2015} \times 0$
$= 0$

7. We have,

$$\sum_{k=0}^{2016} i^k + \sum_{p=0}^{2018} i^p = x + iy$$

$$\Rightarrow x + iy = \sum_{k=0}^{2016} i^k + \sum_{p=0}^{2018} i^p$$

$\Rightarrow x + iy = i^{2016} + i^{2016} + i^{2017} + i^{2018}$

$\Rightarrow x + iy = 1 + 1 + i + i^2 = 1 + i$

Thus, $x = 1, y = 1$

Hence, the value of

$x + y + 2$
$= 1 + 1 + 2$
$= 4$

8. We have,

$(1 + i)^{2n} = (1 - i)^{2n}$

$$\Rightarrow \frac{(1+i)^{2n}}{(1-i)^{2n}} = 1$$

$$\Rightarrow \left(\frac{1+i}{1-i}\right)^{2n} = 1$$

$$\Rightarrow \left(\frac{1+i}{1-i} \times \frac{1+i}{1+i}\right)^{2n} = 1$$

$$\Rightarrow \left(\frac{(1+i)^2}{(1)^2-(i)^2}\right)^{2n} = 1$$

$$\Rightarrow \left(\frac{1+i^2+2i}{1+1}\right)^{2n} = 1$$

$\Rightarrow (i)^{2n} = 1$

$\Rightarrow n = 2$

Hence, the positive integer n is 2.

9. We have

$(1 + i)^5 + (1 + i^3)^5 + (1 + i^5)^7 + (1 + i^7)^7$
$= (1 + i)^5 + (1 - i)^5\} + \{(1 + i)^7 + (1 - i)^7$
$= 21 \pm {}^5C_1 i + {}^5C_1 i^2 \pm {}^5C_1 i^3 + {}^5C_1 i^4 \pm {}^5C_1 i^5\}$
$2(1 \pm {}^7C_1 i + {}^7C_2 i^2 \pm {}^7C_3 i^3 + {}^7C_4 i^4 \pm {}^7C_5 i^5 + {}^7C_6 i^6 \pm {}^7C_7 i^7)$
$= (2 - 20 + 10) + (1 - 21 + 35 - 7)$
$= -8 + 8$
$= 0$

10. We have $z = (n + i)^4$
For $n = 0$, $z = i^4 = 1$, an integer
For $n = 1$, $z = (1 + i)^4 = 1 - 6 + 1 = -4$, an integer
For $n = -1$, $z = (1 + i)^4 = 1 - 6 + 1 = -4$, an integer
For $n = R - \{-1, 0, 1\}$, z, an imaginary number.
Thus, the number of integral values of n is 3.

11. We have,
$$z^2 = (1 + i)^2$$
$$= 1 + i^2 + 2i$$
$$= 2i$$
Hence, the multiplicative inverse of z^2 is $= \frac{1}{2i} = -\frac{i}{2}$.

12 We have $z = \dfrac{1+2i}{3-4i} = \dfrac{(1+2i)(3+4i)}{(3-4i)(3+4i)}$
$$z = \frac{3+4i+6i-8}{25} = \frac{-5+10i}{25}$$
Hence, the multiplicative inverse of z is
$$\frac{25}{-5+10i}$$
$$= \frac{25(-5-10i)}{(-5+10i)(-5-10i)}$$
$$= \frac{25(-5-10i)}{125}$$
$$= \frac{(-5-10i)}{5} = -1-2i$$

13. The given relation $a + ib > c + id$ holds good only when if $b = 0$, $d = 0$.
Thus, $b + d + 2016 = 0 + 0 + 2016 = 2016$.

14. We have,
$$\left(\frac{1+i}{1-i}\right)^n = \frac{2}{\pi}\left(\sin^{-1}x + \sec^{-1}\left(\frac{1}{x}\right)\right)$$
$$\Rightarrow \left(\frac{1+i}{1-i}\right)^n = \frac{2}{\pi}(\sin^{-1}x + \cos^{-1}x)$$
$$\Rightarrow \left(\frac{1+i}{1-i}\right)^n = \frac{2}{\pi} \times \frac{\pi}{2} = 1$$
$$\Rightarrow \left(\frac{1+i}{1-i}\right)^n = 1$$
$$\Rightarrow i^n = 1$$
$$\Rightarrow i^n = i^4$$
$$\Rightarrow n = 4$$
Thus, the positive integer is 4.

15. We have
$$\frac{(1+i)x-2i}{(3+i)} + \frac{(2-3i)y+\mathrm{i}}{(3-i)} = i$$
$$\Rightarrow \frac{1+i(x-2)}{(3+i)} + \frac{2+\mathrm{i}(1-3y)}{(3-i)} = \mathrm{i}$$
$$\Rightarrow (4 + 2i)x + (9 - 7i)y - 3i - 3 = 10i$$
Equating the real and imaginary parts, we get,
$2x - 7y = 13$ and $4x + 9y = 3$.
Hence, $x = 3$, $y = -1$.

16. We have
$$x + iy = \frac{2^{1008}}{(1+i)^{2016}} + \frac{(1+i)^{2016}}{2^{1008}}$$
$$= \left(\frac{2}{(1+i)^2}\right)^{1008} + \left(\frac{(1+\mathrm{i})^2}{2}\right)^{1008}$$
$$= \left(\frac{2}{2i}\right)^{1008} + \left(\frac{2i}{2}\right)^{1008}$$
$$= \left(\frac{1}{i}\right)^{1008} + (i)^{1008}$$
$$= (-i)^{1008} + (i)^{1008}$$
$$= 1 + 1$$
$$= 2$$
Thus, $x = 2$ and $y = 0$.

17. We have
$$z^{1/3} = a + b$$
$$\Rightarrow (x + i\,y)^{1/3} = a + i\,b$$
$$\Rightarrow (x + i\,y) = (a + i\,b)^3$$
$$= \mathrm{a}^3 + 3a^2\,(i\,b) + 3a\,(i\,b)^2 + (i\,b)^3$$
$$\Rightarrow (x + iy) = a^3 + i^3a^2b - 3ab^2 - ib^3$$
$$\Rightarrow (x + iy) = (a^3 - 3ab^2) + i(3a^2b - b^3)$$
$$\Rightarrow x = (a^3 - 3ab^2) \text{ and } y = (3a^2b - b^3)$$
$$\Rightarrow \frac{x}{a} = a^2 - 3b^2 \text{ and } \frac{y}{b} = 3a^2 - b^2$$
$$\Rightarrow \frac{x}{a} + \frac{y}{b} = (a^2 - 3b^2) + (3a^2 - b^2)$$
Adding, we get
$$\Rightarrow \frac{x}{a} + \frac{y}{b} = (4a^2 - 4b^2)$$
$$\Rightarrow \frac{x}{a} + \frac{y}{b} = 4(a^2 - b^2)$$

18. Given
$$x = 3 + 2i$$
$$\Rightarrow (x - 3)^2 = -4$$
$$\Rightarrow x^2 - 6x + 9 = -4$$
$$\Rightarrow x^2 - 6x + 13 = 0$$
We have $x^4 - 8x^3 + 4x^2 + 4x + 39$
$$= x^2(x^2 - 6x + 13) - 2x^3 - 9x^2 + 4x + 39$$
$$= -2x^3 - 9x^2 + 4x + 39$$
$$= -2x(x^2 - 6x + 13) - 21x^2 + 30x + 39$$
$$= -21x^2 + 30x + 39$$
$$= -21(x^2 - 6x + 13) - 96x + 312$$
$$= -96x + 312$$
$$= -96(3 + 2i) + 312$$
$$= -288 - 192i + 312$$
$$= 24 - 192i$$
Thus, $a = 24$ and $b = -192$.

Hence, the value of

$$\left(\frac{b}{a}+10\right)$$

$$=\left(-\frac{192}{24}+10\right)$$

$$= -8 + 10$$

$$= 2$$

19. We have,

$$z=\left(\frac{2i}{1+i}\right)^n$$

$$=\left(\frac{2i}{1+i}\times\frac{(1-i)}{(1-i)}\right)^n$$

$$=\left(\frac{2i(1-i)}{2}\right)^n$$

$$= (1 + i)^n$$

$$=\left(\sqrt{2}\left(\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}\right)\right)^n$$

$$=\left(\sqrt{2}\left(\cos\left(\frac{\pi}{4}\right)+i\sin\left(\frac{\pi}{4}\right)\right)\right)^n$$

$$=(2)^{\frac{n}{2}}\left(\cos\left(\frac{n\pi}{4}\right)+i\sin\left(\frac{n\pi}{4}\right)\right)$$

It will be a real number if

$$\sin\left(\frac{n\pi}{4}\right)=0$$

$$\Rightarrow \quad \left(\frac{n\pi}{4}\right)=k\pi, k=1,2,3\ldots$$

$$\Rightarrow \quad n = 4k, k = 1, 2, 3, \ldots$$

For $n = 4,\ z=4\cos\left(\frac{4\pi}{4}\right)=-4$

and for $n = 8,\ z=16\cos\left(\frac{8\pi}{4}\right)=16$

Hence, the minimum value of n is 8.

20. As we know that, if one root of a quadratic equation is imaginary, its other one will be its conjugate.
Thus, the another root is $3 - 2i$.
Now, the sum of the roots $= 3 + 2i + 3 - 2i$
$= 6$
and product of the roots
$= (3 + 2i)(3 - 2i)$
$= (3)^2 - (2i)^2 = 9 + 4 = 13$
Hence, the required equation is
$x^2 - Sx + P = 0$
$\Rightarrow \quad x^2 - 6x + 13 = 0$

21. We have $z^2 + \bar{z} = 0$.
Let $z = x + iy$
Then $(x + iy)^2 + (x - iy) = 0$
$\Rightarrow \quad (x^2 - y^2 + i2xy) + (x - iy) = 0$
$\Rightarrow \quad (x^2 - y^2 + x) + i(2xy - y) = 0$
$\Rightarrow \quad (x^2 - y^2 + x) = 0, (2xy - y) = 0$
Now, $2xy - y = 0$ gives
$\Rightarrow \quad y(2x - 1) = 0$
$\Rightarrow \quad y = 0, x = 1/2$
When $y = 0$, then
$x^2 + x = 0$
$\Rightarrow \quad x(x + 1) = 0 \quad \Rightarrow \quad x = 0, -1$
So the solutions are $(0, 0), (-1, 0)$.
When $x = 1/2$, then

$$\frac{1}{4}-y^2+\frac{1}{2}=0$$

$$\Rightarrow \quad y^2=\frac{3}{4}$$

$$\Rightarrow \quad y=\pm\frac{\sqrt{3}}{2}$$

So, the solutions are $\left(\frac{1}{2},\frac{\sqrt{3}}{2}\right),\left(\frac{1}{2},-\frac{\sqrt{3}}{2}\right)$.

Hence, the solutions of the given equations are

$(0, 0), (-1, 0), \left(\frac{1}{2},\frac{\sqrt{3}}{2}\right),\left(\frac{1}{2},-\frac{\sqrt{3}}{2}\right)$

i.e. $0, -1, \frac{1}{2}+i\frac{\sqrt{3}}{2},\frac{1}{2}-i\frac{\sqrt{3}}{2}$

22. Let z_1, z_2, z_3 are the roots of the given equation.
Thus, $z_1 + z_2 + z_3 = 2(2 - i)$

$$\Rightarrow \quad z_2 + z_3 = 2(2 - i) - z_1$$
$$= 2(2 - i) - (1 - i)$$
$$= 3 - i \qquad \ldots(i)$$

Also, $z_1.z_2.z_3 = (1 - 3i)$

$$z_2\cdot z_3=\frac{1-3i}{1-i}=\frac{(1-3i)(1+i)}{(1-i)(1+i)}$$

$$\Rightarrow \quad =\frac{1+i-3i+3}{2}$$

$$=\frac{4-2i}{2}=2-i \qquad \ldots(ii)$$

Solving Eqs (i) and (ii), we get
$z_1 = 1, z_2 = 2 - i$

23. If one of the given equation is $1 + 2i$, then its other root will be its conjugate. Thus $1 - 2i$ is the other root. Now,
$S = 1 + 2i + 1 - 2i$
$= 2$
and $P = (1 + 2i)(1 - 2i)$
$= 5$
Therefore, $(x^2 - 2x + 5)$ is a factor of the given equation.
The given equation can also be written as
$x^2(x^2 - 2x + 5) - x(x^2 - 2x + 5) + (x^2 - 2x + 5) = 0$
$\Rightarrow \quad (x^2 - 2x + 5)(x^2 - x + 1) = 0$
$\Rightarrow \quad (x^2 - 2x + 5) = 0, (x^2 - x + 1) = 0$

$$\Rightarrow \quad x=1\pm 2i, x=\left(\frac{1}{2}-i\frac{\sqrt{3}}{2}\right),\left(\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)$$

Hence, the solutions are

$$\left\{1 \pm 2i, \left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right), \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)\right\}$$

24. Thus, the greatest and the least value of $|z|$ are $\sqrt{13}+1, \sqrt{13}-1$ respectively.

25. Thus, the greatest and the least value of $|z|$ are $\sqrt{34}+2, \sqrt{34}-2$.

Hence, the required difference

$$= (\sqrt{34}+2) - (\sqrt{34}-2)$$
$$= 4$$

26. Given $a + ib = (1 + i)(1 + 2i)(1 + 3i) \ldots (1 + ni)$

$$\Rightarrow \quad |a + ib| = |(1 + i)(1 + 2i)(1 + 3i) \ldots (1 + ni)|$$
$$= |(1 + i)||(1 + 2i)||(1 + 3i)| \ldots |(1 + ni)|$$
$$\Rightarrow \quad \sqrt{a^2+b^2} = \sqrt{1^2+1^2}\sqrt{1^2+2^2}\sqrt{1^2+3^2}\ldots\sqrt{1^2+n^2}$$
$$\Rightarrow \quad (a^2 + b^2) = 2.5.10 \ldots (n^2 + 1)$$

Hence, the result.

27. We have,

$$x + iy = \frac{a+ib}{a-ib}$$
$$\Rightarrow \quad |x+iy| = \left|\frac{a+ib}{a-ib}\right|$$
$$\Rightarrow \quad = \frac{|a+ib|}{|a-ib|}$$
$$\Rightarrow \quad |x+iy|^2 = \frac{|a+ib|^2}{|a-ib|^2}$$
$$\Rightarrow \quad x^2+y^2 = \frac{(a^2+b^2)}{(a^2+b^2)} = 1$$

Hence, the result.

28. We have,

$$z + |z| = 2 + 8i$$
$$\Rightarrow \quad (x+iy) + \sqrt{x^2+y^2} = 2+8i$$
$$\Rightarrow \quad (x+\sqrt{x^2+y^2}) + iy = 2+8i$$

Thus, $x + \sqrt{x^2+y^2} = 2, y = 8$

$$\Rightarrow \quad (x+\sqrt{x^2+64}) = 2$$
$$\Rightarrow \quad \sqrt{x^2+64} = (2-x)$$
$$\Rightarrow \quad (x^2 + 64) = (2 - x)^2$$
$$\Rightarrow \quad = 4 - 4x + x^2$$
$$\Rightarrow \quad 4x = -60$$
$$\Rightarrow \quad x = -15$$

Hence, $|z| = \sqrt{x^2+y^2}$

$$= \sqrt{225+64} = \sqrt{289} = 17$$

29. We have,

$$iz = ire^{i\theta} = e^{i\frac{\pi}{2}}re^{i\theta} = re^{i\left(\frac{\pi}{2}+\theta\right)}$$
$$= r\left(\cos\left(\frac{\pi}{2}+\theta\right) + i\sin\left(\frac{\pi}{2}+\theta\right)\right)$$
$$= r(-\sin\theta + i\cos\theta)$$

Now, $e^{iz} = e^{r(-\sin\theta + i\cos\theta)} = e^{-r\sin\theta + i(r\cos\theta)}$

$$= e^{-r\sin\theta}e^{-i(r\cos\theta)}$$
$$\Rightarrow \quad |e^{iz}| = |e^{-r\sin\theta}e^{-i(r\cos\theta)}|$$
$$= e^{-r\sin\theta}|e^{-i(r\cos\theta)}|$$
$$= e^{-r\sin\theta} \times 1$$
$$= e^{-r\sin\theta}$$

30. We have,

$$\left|\frac{\alpha\bar{\beta}+\beta\bar{\alpha}}{|\alpha\beta|}\right|$$
$$= \left|\frac{|\alpha\bar{\beta}+\beta\bar{\alpha}|}{|\alpha\beta|}\right|$$
$$\le \frac{|\alpha\bar{\beta}|+|\beta\bar{\alpha}|}{|\alpha\beta|}$$
$$= \frac{|\alpha||\bar{\beta}|+|\beta||\bar{\alpha}|}{|\alpha||\beta|}$$
$$= \frac{|\alpha||\bar{\beta}|}{|\alpha||\beta|} + \frac{|\beta||\bar{\alpha}|}{|\alpha||\beta|}$$
$$= 2 \qquad (\because |\alpha| = |\bar{\alpha}|, |\beta| = |\bar{\beta}|)$$

Thus, the maximum value is 2.

31. Given $|z_1| = 1$

$$\Rightarrow \quad |z_1|^2 = 1 \quad \Rightarrow z_1 \cdot \bar{z}_1 = 1$$
$$\Rightarrow \quad \bar{z}_1 = \frac{1}{z_1}$$

Similarly, $\bar{z}_2 = \frac{1}{z_2}, \bar{z}_3 = \frac{1}{z_3}$

Now, $\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right| = 1$

$$\Rightarrow \quad |\bar{z}_1 + \bar{z}_2 + \bar{z}_3| = 1$$
$$\Rightarrow \quad |\overline{z_1 + z_2 + z_3}| = 1$$
$$\Rightarrow \quad |z_1 + z_2 + z_3| = 1$$

32. We have $z = (3 + 7i)(p + iq), p, q \in I - \{0\}$

$$\Rightarrow \quad z = (3p - 7q) + i(7p + 3q)$$
$$\Rightarrow \quad |z| = \sqrt{(3p-7q)^2+(7p+3q)^2}$$
$$\Rightarrow \quad |z|^2 = (3p - 7q)^2 + (7p + 3q)^2$$
$$= 9p^2 + 49q^2 + 49p^2 + 9q^2$$
$$= 58(p^2 + q^2) \qquad \ldots(i)$$

Since z is purely imaginary number, so

$$(3p - 7q) = 0$$

$\Rightarrow \quad \frac{p}{7} = \frac{q}{3} = \lambda(\text{say}), \lambda \in I - \{0\}$

Thus, $|z|^2$ will be minimum only when $\lambda = 1$.
Therefore, $p = 7$ and $q = 3$
Hence, the minimum value of $|z|^2 = 58(49 + 9)$
$= 58 \times 58$
$= 3364$

33. Given,
$|\beta| = 1$
$\Rightarrow \quad |\beta|^2 = 1$
$\Rightarrow \quad \beta \cdot \bar{\beta} = 1$
$\Rightarrow \quad \beta = \frac{1}{\bar{\beta}}$

Now, $\left|\frac{\beta - \alpha}{1 - \bar{\alpha}\beta}\right| = \left|\frac{\beta - \alpha}{1 - \bar{\alpha} \cdot \frac{1}{\bar{\beta}}}\right|$

$= \left|\frac{\bar{\beta}(\beta - \alpha)}{\bar{\beta} - \bar{\alpha}}\right|$

$= \frac{|\bar{\beta}||\beta - \alpha|}{|\bar{\beta} - \bar{\alpha}|}$

$= \frac{|\bar{\beta}||\beta - \alpha|}{|\overline{(\beta - \alpha)}|} = 1$

34. We have $|z + 1| = z + 2(1 + i)$.
Let $z = x + iy$.
Then $|x + iy + 1| = (x + iy) + 2(1 + i)$
$\Rightarrow \quad |(x + 1) + iy| = (x + 2) + i(y + 2)$
$\Rightarrow \quad \sqrt{(x+1)^2 + y^2} = (x + 2) + i(y + 2)$
Comparing the real and imaginary parts, we get
$\sqrt{(x+1)^2 + y^2} = (x + 2), (y + 2) = 0$
when $y = -2, x = 1/2$
Hence, the complex number is
$z = x + iy = \frac{1}{2} - 2i$

35. We have
$|z - 1|^2 + |z + 1|^2 = 5$
$\Rightarrow \quad |(x + iy) - 1|^2 + |(x + iy) + 1|^2 = 5$
$\Rightarrow \quad |(x - 1) - iy|^2 + |(x + 1) + iy|^2 = 5$
$\Rightarrow \quad (x - 1)^2 + y^2 + (x + 1)^2 + y^2 = 8$
$\Rightarrow \quad 2(x^2 + y^2) + 2 = 8$
$\Rightarrow \quad 2(x^2 + y^2) = 6$
$\Rightarrow \quad (x^2 + y^2) = 3$
$\Rightarrow \quad |z|^2 = 3$

36. Given,
$|z - 2| = 2|z - 1|$
$\Rightarrow \quad |z - 2|^2 = 4|z - 1|^2$
$\Rightarrow \quad (z - 2)(\bar{z}_1 - 2) = 4(z - 1)(\bar{z}_1 - 1)$
$\Rightarrow \quad (z \cdot \bar{z} - 2z - 2\bar{z} + 4) = 4(z \cdot \bar{z} - z - \bar{z} + 1$
$\Rightarrow \quad (|z|^2 - 2z - 2\bar{z} + 4) = 4(|z|^2 - z - \bar{z} + 1)$
$\Rightarrow \quad 3|z|^2 - 2z - 2\bar{z} = 0$
$\Rightarrow \quad 3|z|^2 - 2(z - 2\bar{z}) = 0$
$\Rightarrow \quad 3|z|^2 - 2(z - 2\bar{z}) = 2.2\text{Re}(z) = 4\text{Re}(z)$
$\Rightarrow \quad |z|^2 = \frac{4}{3}\text{Re}(z)$
Hence, the result.

37. Given,
$|z + 6| = |3z + 2|$
$\Rightarrow \quad |z + 6|^2 = |3z + 2|^2$
$\Rightarrow \quad (z + 6)(\bar{z} + 6) = (3z + 2)(3\bar{z} + 2)$
$\Rightarrow \quad (z.\bar{z} + 6(z + \bar{z}) + 36) = (9z.\bar{z} + 6(z + \bar{z}) + 4)$
$\Rightarrow \quad (z.\bar{z} + 36) = (9z.\bar{z} + 4)$
$\Rightarrow \quad (|z|^2 + 36) = (9|z|^2 + 4)$
$\Rightarrow \quad 8|z|^2 = 32$
$\Rightarrow \quad |z|^2 = 4$
$\Rightarrow \quad |z| = 2$
Hence, the value of $|z|$ is 2.

38. Given,
$|z + 6| = |2z + 3|$
$\Rightarrow \quad |x + iy + 6| = |2(x + iy) + 3|$
$\Rightarrow \quad |(x + 6) + iy| = |(2x + 3) + i.2y|$
$\Rightarrow \quad |(x + 6) + iy|^2 = |(2x + 3) + i.2y|^2$
$\Rightarrow \quad (x + 6)^2 + y^2 = (2x + 3)^2 + 4y^2$
$\Rightarrow \quad x^2 + 12x + 36 + y^2 = 4x^2 + 12x + 9 + 4y^2$
$\Rightarrow \quad 3x^2 + 3y^2 = 27$
$\Rightarrow \quad x^2 + y^2 = 9$
Hence, the locus of z is $x^2 + y^2 = 9$.

39. Given,
$|z| = 1$
$\Rightarrow \quad |z|^2 = 1$
$\Rightarrow \quad z.\bar{z} = 1$
$\Rightarrow \quad \bar{z} = \frac{1}{z}$

Now, $2\text{Re}(\omega) = (\omega + \bar{\omega})$

$= \left(\frac{z-1}{z+1}\right) + \overline{\left(\frac{z-1}{z+1}\right)}$

$= \left(\frac{z-1}{z+1}\right) + \left(\frac{\bar{z}-1}{\bar{z}+1}\right)$

$= \left(\frac{z-1}{z+1}\right) + \left(\frac{(1/z)-1}{(1/z)+1}\right)$

$= \left(\frac{z-1}{z+1}\right) + \left(\frac{1-z}{1+z}\right)$

$= \left(\frac{z-1+1-z}{z+1}\right)$

$= 0$

$\Rightarrow \quad \text{Re}(\omega) = 0$

40. We have,
$|z + i| + |z - i| = 8$

$\Rightarrow \quad 8 = |z+i| + |z-i|$

$\Rightarrow \quad 8 = |z+i| + |z-i| \geq |z+i+z-i|$

$\Rightarrow \quad |z+i+z-i| \leq 8$

$\Rightarrow \quad |2z| \leq 8$

$\Rightarrow \quad |z| \leq 4$

Hence, the maximum value of $|z|$ is 4.

41. (i) Let $z = 1 + i = (1, 1)$.

We have $\alpha = \tan^{-1}\left|\frac{1}{1}\right| = \tan^{-1}(1) = \frac{\pi}{4}$

Since, the given complex number lies in the first first quadrant, so $\text{Arg}(z) = \theta = \alpha = \frac{\pi}{4}$

(ii) Let $z = 1 - i = (1, -1)$.

We have $\alpha = \tan^{-1}\left|\frac{-1}{1}\right| = \tan^{-1}(1) = \frac{\pi}{4}$

Since the complex number z lies in the fourth quadrant, so

$\text{Arg}(z) = \theta = -\alpha = -\frac{\pi}{4}$

(iii) Let $z = -1 + i = (-1, 1)$

We have $\alpha = \tan^{-1}\left|\frac{1}{-1}\right| = \tan^{-1}(1) = \frac{\pi}{4}$

Since the complex number lies in the second quadrant, so

$$\text{Arg}(z) = \theta = \pi - \alpha = \pi - \frac{\pi}{4} = \frac{3\pi}{4}.$$

(iv) Let $z = -1 - i = (-1, -1)$

We have $\alpha = \tan^{-1}\left|\frac{-1}{-1}\right| = \tan^{-1}(1) = \frac{\pi}{4}$

Since the complex number z lies in the third quadrant, so

$\text{Arg}(z) = \theta = -(\pi - \alpha)$

42. Now $-z = -\frac{\sqrt{3}}{2} - \frac{i}{2} = \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

We have $\alpha = \tan^{-1}\left|\frac{-1/2}{-\sqrt{3}/2}\right| = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

Since the complex number $-z$ lies in the third quadrant, so

$\text{Arg}(-z) = \theta = -(\pi - \alpha)$

$= -\left(\pi - \frac{\pi}{6}\right) = -\frac{5\pi}{6}$

$= -\left(\pi - \frac{\pi}{4}\right) = -\frac{3\pi}{4}$

Similarly, you can solve the other parts easily.

43. Since $\text{Arg}(z) < 0$, so z lies in the third quadrant and $(-z)$ lies in the first quadrant

Thus,

$\text{Arg}(z) - \text{Arg}(-z) = -(\pi - \alpha) - \theta = -\pi$

44. Given $|z+1| = |z-1|$

$\Rightarrow \quad |x + iy + 1| = |x + iy - 1|$

$\Rightarrow \quad |(x+1) + iy| = |(x-1) + iy|$

$\Rightarrow \quad \sqrt{(x+1)^2 + y^2} = \sqrt{(x-1)^2 + y^2}$

$\Rightarrow \quad (x+1)^2 + y^2 = (x-1)^2 + y^2$

$\Rightarrow \quad 4x = 0 \Rightarrow x = 0$

Also, $\text{Amp}\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$

$\Rightarrow \quad \text{Amp}(z-1) - \text{Amp}(z+1) = \frac{\pi}{4}$

$\Rightarrow \quad \tan^{-1}\left(\frac{y}{x-1}\right) - \tan^{-1}\left(\frac{y}{x+1}\right) = \frac{\pi}{4}$

$\Rightarrow \quad \tan^{-1}\left(\frac{\frac{y}{x-1} - \frac{y}{x+1}}{1 + \frac{y}{x-1} \cdot \frac{y}{x+1}}\right) = \frac{\pi}{4}$

$\Rightarrow \quad \tan^{-1}\left(\frac{xy + y - xy + y}{x^2 + y^2 - 1}\right) = \frac{\pi}{4}$

$\Rightarrow \quad \tan^{-1}\left(\frac{2y}{x^2 + y^2 - 1}\right) = \frac{\pi}{4}$

$\Rightarrow \quad \frac{2y}{x^2 + y^2 - 1} = 1$

$\Rightarrow \quad x^2 + y^2 - 1 = 2y$

$\Rightarrow \quad x^2 + y^2 - 2y - 1 = 0$

$\Rightarrow \quad y^2 - 2y - 1 = 0$, since $x = 0$

$\Rightarrow \quad y = \frac{2 \pm \sqrt{8}}{2} = (1 \pm \sqrt{2})$

Thus, the complex number,

$z = x + iy = i(1 \pm \sqrt{2})$

45. Given,

$|z_1 + z_2| = |z_1 - z_2|$

$\Rightarrow \quad |z_1 + z_2|^2 = |z_1 - z_2|^2$

$\Rightarrow \quad |z_1|^2 + |z_1|^2 + 2\text{Re}(z_1\bar{z}_2) = |z_1|^2 + |z_1|^2 + 2\text{Re}(z_1\bar{z}_2)$

$\Rightarrow \quad 4\text{Re}(z_1\bar{z}_2) = 0$

$\Rightarrow \quad \text{Re}(e^{i(\theta_1 + \theta_2)}) = 0$

$\Rightarrow \quad \text{Re}(\cos(\theta_1 - \theta_2) - i(\theta_1 - \theta_2)) = 0$

$\Rightarrow \quad \cos(\theta_1 - \theta_2) = 0$

$\Rightarrow \quad \theta_1 - \theta_2 = \frac{\pi}{2}$

Thus, $\text{Amp}\left(\frac{z_1}{z_2}\right)(\theta_1 - \theta_2) = \frac{\pi}{2}$

46. We have,

$$z = \left(\frac{\cos\theta + i\sin\theta}{\cos\theta - i\sin\theta}\right)$$

$$= (\cos 2\theta + i\sin 2\theta)$$

Also, it is given that $\frac{\pi}{4} < \theta < \frac{\pi}{2}$

$\Rightarrow \quad \frac{\pi}{2} < 2\theta < \pi$

Thus, Arg $(z) = \pi - 2\theta$

47. We have,

$$z = \sin\left(\frac{\pi}{5}\right) + i\left(1 - \cos\left(\frac{\pi}{5}\right)\right)$$

$$= \sin\left(\frac{\pi}{5}\right) + \mathrm{i}\left(1 - \cos\left(\frac{\pi}{5}\right)\right)$$

$$= \sin\left(\frac{\pi}{5}\right) + 2 \cdot i \sin^2\left(\frac{\pi}{10}\right)$$

Thus, $\text{Arg}(z) = \tan^{-1}\left(\frac{2\sin^2(\pi/10)}{\sin(\pi/5)}\right)$

$$= \tan^{-1}\left(\frac{2\sin^2(\pi/10)}{2\sin(\pi/10)\cos(\pi/10)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{10}\right)\right)$$

$$= \frac{\pi}{10}$$

48. Let $z = x + iy$.
Given,

$$\text{Arg}(z) = \frac{\pi}{3}$$

$\Rightarrow \quad \tan^{-1}\left(\frac{y}{x}\right) = \frac{\pi}{3}$

$\Rightarrow \quad \frac{y}{x} = \tan\left(\frac{\pi}{3}\right) = \sqrt{3}$

$\Rightarrow \quad y = x\sqrt{3}$

Now, $z - 1 = (x + iy) - 1 = (x - 1) + iy$

Also, $\text{Arg}(z - 1) = \frac{5\pi}{6}$

$\Rightarrow \quad \tan^{-1}\left(\frac{y}{x-1}\right) = \frac{5\pi}{6}$

$\Rightarrow \quad \frac{y}{x-1} = \tan\left(\frac{5\pi}{6}\right) = -\sqrt{3}$

$\Rightarrow \quad x\sqrt{3} = -\sqrt{3}(x - 1)$

$\Rightarrow \quad x = -x + 1$

$\Rightarrow \quad 2x = 1$

$\Rightarrow \quad x = \frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$

Thus, the complex number is $\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$.

49. Let $z = e^{i\theta} = \cos\theta + i\sin\theta$

We have $|z - 3i| = 3$

$\Rightarrow \quad |\cos\theta + i(\sin\theta - 3)| = 3$

$\Rightarrow \quad \sqrt{\cos^2\theta + (\sin\theta - 3)^2} = 3$

$\Rightarrow \quad \cos^2\theta + (\sin\theta - 3)^2 = 9$

$\Rightarrow \quad \cos^2\theta + \sin^2\theta - 6\sin\theta + 9 = 9$

$\Rightarrow \quad 1 - 6\sin\theta = 0$

$\Rightarrow \quad \sin\theta = \frac{1}{6}$ (i)

Now, $\cot\theta - \frac{6}{z} = \cot\theta - \frac{6}{(\cos\theta + i\sin\theta)}$

$$= \cot\theta - 6(\cos\theta - i\sin\theta)$$

$$= \cot\theta - \frac{1}{\sin\theta}(\cos\theta - i\sin\theta)$$

$$= \cot\theta - \cot\theta + i$$

$$= i$$

Therefore,

$$\text{Arg}\left(\cot\theta - \frac{6}{z}\right) = \text{Arg}(i) = \frac{\pi}{2}$$

50. We have,

$$\text{Amp}\left(\frac{z-1}{z+1}\right) = \frac{\pi}{3}$$

$\Rightarrow \quad \text{Amp}\left(\frac{x + iy - 1}{x + iy + 1}\right) = \frac{\pi}{3}$

$\Rightarrow \quad \text{Amp}\left(\frac{(x-1) + iy}{(x+1) + iy}\right) = \frac{\pi}{3}$

$\Rightarrow \quad \text{Amp}\left(\frac{(x-1) + iy}{(x+1) + iy} \times \frac{(x+1) - iy}{(x+1) - iy}\right) = \frac{\pi}{3}$

$\Rightarrow \quad \text{Amp}\left(\frac{(x^2 - 1) + y^2 + i(xy + y - xy + y)}{(x+1)^2 + y^2}\right) = \frac{\pi}{3}$

$\Rightarrow \quad \text{Amp}\left(\frac{(x^2 + y^2 - 1)}{(x+1)^2 + y^2} + i\frac{(2y)}{(x+1)^2 + y^2}\right) = \frac{\pi}{3}$

$\Rightarrow \quad \tan^{-1}\left(\frac{2y}{(x^2 + y^2 - 1)}\right) = \frac{\pi}{3}$

$\Rightarrow \quad \frac{2y}{(x^2 + y^2 - 1)} = \tan\left(\frac{\pi}{3}\right)$

$\Rightarrow \quad \frac{2y}{(x^2 + y^2 - 1)} = \sqrt{3}$

$\Rightarrow \quad x^2 + y^2 - \frac{2}{\sqrt{3}}y - 1 = 0$

Hence, the locus of z is a circle.

51. We have $\frac{1}{(\sqrt{3} - i)^{25}}$.

Let $z = \left(\frac{\sqrt{3} + i}{4}\right)$

Then $\text{Arg}(z) = \tan^{-1}\left(\frac{1/4}{\sqrt{3}/4}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

$\Rightarrow \quad \text{Arg}(z) = \dfrac{\pi}{6}$

Hence, the angle is $\dfrac{\pi}{6}$.

52. We have, $|z^1 + z_2|^2$

$= r_1(\cos\theta_1 + i\sin\theta_2) + r_2(\cos\theta_2 + i\sin\theta_2)|^2$

$= r_1|(\cos\theta_1 + r_2\cos\theta_2) + i(r_2\sin\theta_1 + r_2\sin\theta_2)|^2$

$= r_1(\cos\theta_1 + r_2\cos\theta_2)^2 + (r_1\sin\theta_1 + r_2\sin\theta_2)^2$

$= r_1^2 + r_2^2 + 2r_1r_2\cos(\theta_1 - \theta_2)$

$\leq r_1^2 + r_2^2 + 2r_1r_2$

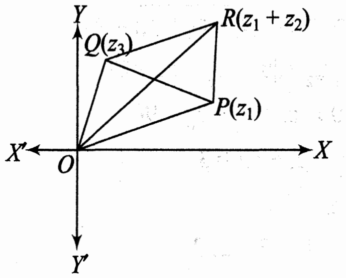
$= (r_1 + r_2)^2$

$= (|z_1| + |z_2|)^2$

$\Rightarrow \quad |z_1 + z_2| \leq |z_1| + |z_2|$

Hence, the result.

Alternate method:



In ΔOPR, $OP + PR \geq OR$

$\Rightarrow \quad |z_1| + |z_2| \geq |z_1 + z_2|$

$\Rightarrow \quad |z_1| + |z_2| \leq |z_1| + |z_2|$

53. In ΔOPQ, $|OP - OQ| \leq PQ$

$\Rightarrow \quad ||z_1| - |z_2|| \leq |z_1 - z_2|$

Hence, the result.

54. We have,

$|z_1 + z_2| = |z_1 - z_2|$

$\Rightarrow \quad |z_1 + z_2|^2 = |z_1 - z_2|^2$

$\Rightarrow \quad |z_1|^2 + |z_2|^2 + 2|z_1||z_2|\cos(\theta_1 - \theta_2)$

$= |z_1|^2 + |z_2|^2 - 2|z_1||z_2|\cos(\theta_1 - \theta_2)$

$\Rightarrow \quad 4|z_1||z_2|\cos(\theta_1 - \theta_2) = 0$

$\Rightarrow \quad \cos(\theta_1 - \theta_2) = 0$

$\Rightarrow \quad (\theta_1 - \theta_2) = \dfrac{\pi}{2}$

$\Rightarrow \quad \text{Arg}(z_1) - \text{Arg}(z_2) = \dfrac{\pi}{2}$

55. We have,

$|z_1 + z_2| = |z_1| + |z_2|$

$\Rightarrow \quad |z_1 + z_2|^2 = (|z_1| + |z_2|)^2$

$\Rightarrow \quad |z_1|^2 + |z_2|^2 + 2|z_1||z_2|\cos(\theta_1 - \theta_2)$

$= (|z_1|^2 + |z_2|^2 + 2|z_1||z_2|)$

$\Rightarrow \quad 2|z_1||z_2|\cos(\theta_1 - \theta_2) = 2|z_1||z_2|$

$\Rightarrow \quad \cos(\theta_1 - \theta_2) = 1 = \cos(0)$

$\Rightarrow \quad \theta_1 = \theta_2$

$\Rightarrow \quad \text{Arg}(z_1) = \text{Arg}(z_2)$

Hence, the result.

56. This is possible only when $0, z_1, iz_2$ are on the same side of a straight line.

Thus, $\text{Arg}(z_1) = \text{Arg}(iz_2)$

$\Rightarrow \quad \text{Arg}(z_1) = \text{Arg}(i) + \text{Arg}(iz_2)$

$\Rightarrow \quad \text{Arg}(z_1) = \dfrac{\pi}{2} + \text{Arg}(z_2)$

$\Rightarrow \quad \text{Arg}(z_1) - \text{Arg}(z_2) = \dfrac{\pi}{2}$

Hence, the result.

57. Given,

$|z_1 - z_2| = |z_1| + |z_2|$

$\Rightarrow \quad |z_1 + z_2|^2 = (|z_1| + |z_2|)^2$

$\Rightarrow \quad |z_1|^2 + |z_2|^2 - 2|z_1||z_2|\cos(\theta_1 - \theta_2)$

$= |z_1|^2 + |z_2|^2 - 2|z_1||z_2|$

$\Rightarrow \quad \cos(\theta_1 - \theta_2) = -1 = \cos(\pi)$

$\Rightarrow \quad \theta_1 - \theta_2 = \pi$

$\Rightarrow \quad \text{Arg}(z_1) - \text{Arg}(z_2) = \pi$

$\Rightarrow \quad \text{Arg}\left(\dfrac{z_1}{z_2}\right) = \pi$

$\Rightarrow \quad \left(\dfrac{z_1}{z_2}\right) = -k, k \in I^+$

$\Rightarrow \quad z_1 = -kz_2$

$\Rightarrow \quad z_1 + kz_2 = 0$

Hence, the result.

58. Given,

$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$

$\Rightarrow \quad |z_1|^2 + |z_2|^2 + 2|z_1||z_2|\cos(\theta_1 - \theta_2) = |z_1|^2 + |z_2|^2$

$\Rightarrow \quad \cos(\theta_1 - \theta_2) = 0$

$\Rightarrow \quad \theta_1 - \theta_2 = \dfrac{\pi}{2}$

Now $\dfrac{z_1}{z_2} = \dfrac{|z_1|}{|z_2|}(\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2))$

$= i\dfrac{|z_1|}{|z_2|}$

$\Rightarrow \quad \dfrac{z_1}{z_2}$ is purely an imaginary number.

59. We have $|z_1 - z_2|^2$

$= |z_1|^2 + |z_1|^2 - 2|z_1||z_2|\cos(\theta_1 - \theta_2)$

$= r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_1 - \theta_2)$

$= r_1^2 + r_2^2 - 2r_1r_2 + 2r_1r_2 - 2r_1r_2\cos(\theta_1 - \theta_2)$

$= (r_1 - r_2)^2 + 2r_1r_2(1 - \cos(\theta_1 - \theta_2))$

$$= (r_1 - r_2)^2 + 2r_1r_2 \cdot 2\sin^2\left(\frac{\theta_1 - \theta_2}{2}\right)$$
$$\leq (r_1 - r_2)^2 + 2r_1r_2 \cdot 2 \cdot \left(\frac{\theta_1 - \theta_2}{2}\right)^2$$
$$= (r_1 - r_2)^2 + r_1r_2(\theta_1 - \theta_2)^2$$
$$\leq (r_1 - r_2)^2 + (\theta_1 - \theta_2)^2$$
$$\Rightarrow \quad |z_1 - z_2|^2 \leq (|z_1|^2 - |z_2|^2) + (\text{Arg}(z_1) - \text{Arg}(z_2))^2$$

Hence, the result.

60. Do yourself.

61. We have,

$$|z + 1| = |z + 4 - 3|$$
$$= |(z + 4) + (-3)|$$
$$\leq |z + 4| + |-3|$$
$$\leq 3 + 3 = 6$$

Hence, the maximum value of $|z + 1|$ is 6.

62. We have,

$$|z + 2| = |(z + 5) - 3|$$
$$= |(z + 5) + (-3)|$$
$$\geq |(z + 5) + |-3| = 4 - 3 = 1$$

Hence, the minimum value of $|z + 2|$ is 1.

63. (i) We have,

$$|z| + |z + 2| \geq |z - (z + 2)| = 2$$

Hence, the minimum value is 2.

(ii) We have,

$$|z + 2| + |z - 2| \geq |(z + 2) - (z - 2)| = 4$$

Hence, the minimum value is 4

64. We have,

$$|z + 2| + |z - 2| + |2z - 7|$$
$$= |z + 2| + |z - 2| + |7 - 2z|$$
$$\leq |z + 2 + z - 2 + 2z - 7|$$
$$= 7$$

Hence, the maximum value is 7.

65. We have,

$$|z_1 + z_2| \leq |z_1| + |z_2| = 10 + 5 = 15$$

Also,

$$|z_1 + z_2| \geq |z_1| - |z_2| = 10 - 5 = 5$$

Thus, the greatest value is 15 and the least value is 5.

66. We have,

$$|i(z + 1) + 1| = |i(z - 3i) + i - 2|$$
$$= |i(z - 3i) + (i - 2)|$$
$$\leq |i(z - 3i)| + |(i - 2)|$$
$$\leq 4 + \sqrt{5}$$

Hence, the maximum value is $4 + \sqrt{5}$.

67. We have,

$$\left|z + \frac{1}{z}\right| \geq |z| - \frac{1}{|z|} = 3 - \frac{1}{3} = \frac{8}{3}$$

Hence, the minimum value is $\frac{8}{3}$.

Also,

$$\left|z + \frac{1}{z}\right| \leq |z| + \frac{1}{|z|} = 3 + \frac{1}{3} = \frac{10}{3}$$

Hence, the maximum value is $\frac{10}{3}$.

68. Given $|z_1| = 2, |z_2| = 3, |z_3| = 5$

$$\Rightarrow \quad |z_1|^2 = 4, |z_2|^2 = 9, |z_3|^2 = 25$$
$$\Rightarrow \quad z_1\overline{z_1} = 4, z_2\overline{z_2} = 9, z_3\overline{z_3} = 25$$
$$\Rightarrow \quad \overline{z_1} = \frac{4}{z_1}, \overline{z_2} = \frac{9}{z_2}, \overline{z_3} = \frac{25}{z_3}$$

Also, $|25z_1z_2 + 9z_1z_3 + 4z_3z_2| = 90$

$$\Rightarrow \quad |25z_1z_2 + 9z_1z_3 + 4z_3z_2| = 90$$
$$\Rightarrow \quad \left|z_1z_2z_3\left(\frac{25}{z_3} + \frac{9}{z_2} + \frac{4}{z_1}\right)\right| = 90$$
$$\Rightarrow \quad |z_1z_2z_3|\left|\left(\frac{25}{z_3} + \frac{9}{z_2} + \frac{4}{z_1}\right)\right| = 90$$
$$\Rightarrow \quad |z_1||z_2||z_3||\overline{z_3} + \overline{z_2} + \overline{z_1}| = 90$$
$$\Rightarrow \quad |z_1||z_2||z_3||\overline{z_1 + z_2 + z_3}| = 90$$
$$\Rightarrow \quad |z_1||z_2||z_3||z_1 + z_2 + z_3| = 90$$
$$\Rightarrow \quad 30 \times |z_1 + z_2 + z_3| = 90$$
$$\Rightarrow \quad |z_1 + z_2 + z_3| = 3$$

69. We have,

$$3 - 4i = 3 - 2.\,2.i$$
$$= (2)2 + i^2 - 2.2.i$$
$$= (2 - i)^2$$

Thus, $\sqrt{3 - 4i} = \pm(2 - i)$

70. We have,

$$5 + 12i = 5 + 2.\,3.\,2i$$
$$= (3)^2 + (2i)^2 + 2.3.2i$$
$$= (3 + 2i)^2$$

Thus, $\sqrt{(5 + 12i)} = \pm(3 + 2i)$

71. We have,

$$8 - 6i = 8 - 2.\,3.i$$
$$= (3)^2 + (2i)^2 - 2.3.2i$$
$$= (3 - i)^2$$

Thus, $\sqrt{8 - 6i} = \pm(3 - i)$

72. We have,

$$3i = \frac{3}{2}(2i)$$
$$= \frac{3}{2}(2 \cdot 1 \cdot i)$$
$$= \frac{3}{2}(1^2 + i^2 + 2 \cdot 1 \cdot i)$$
$$= \frac{3}{2}(1 + i)^2$$

Thus, $\sqrt{3i} = \pm\left(\frac{3}{2}(1+i)\right)$

73. We have,

$$8-15i = \frac{1}{2}(16-30i)$$
$$= \frac{1}{2}(16-2\cdot 5\cdot 3i)$$
$$= \frac{1}{2}[5^2+(3i)^2-2\cdot 5\cdot 3i]$$
$$= \frac{1}{2}(5-3i)^2$$

Thus, $\sqrt{8-15i} = \pm\frac{1}{\sqrt{2}}(5-3i)$

74. We have,

$$x^2+\frac{1}{x^2}+4i\left(x-\frac{1}{x}\right)-6$$
$$= \left(x-\frac{1}{x}\right)^2+2\cdot x\cdot\frac{1}{x}+4i\left(x-\frac{1}{x}\right)-6$$
$$= \left(x-\frac{1}{x}\right)^2+4i\left(x-\frac{1}{x}\right)-4$$
$$= \left(x-\frac{1}{x}\right)^2+2\cdot\left(x-\frac{1}{x}\right)\cdot 2i-4$$
$$= \left(x-\frac{1}{x}\right)^2+2\cdot\left(x-\frac{1}{x}\right)\cdot 2i+(2i)^2$$
$$= \left(x-\frac{1}{x}+2i\right)^2$$

Thus, $\sqrt{x^2+\frac{1}{x^2}+4i\left(x-\frac{1}{x}\right)-6}$
$$= \pm\left(x-\frac{1}{x}+2i\right)$$

75. We have,

$z^2+5 = 12\sqrt{-1} = 12i$

$\Rightarrow$ $z^2 = -5+12i$

$\Rightarrow$ $z^2 = -5+2.2.3i$

$\Rightarrow$ $z^2 = (2+3i)^2$

$\Rightarrow$ $z = \pm(2+3i)$

Thus, the complex number z can be $(2+3i)$ or $(-2-3i)$.

76. We have,

$$(2+3\omega+3\omega^2)^{2013} = (2+3(\omega+\omega^2))^{2013}$$
$$= (2+3(-1))^{2013}$$
$$= (-1)^{2013}$$
$$= -1$$

77. We have,

$$(3+4\omega+5\omega^2)^{10} = (3+4(\omega+\omega^2)+\omega^2)^{10}$$
$$= (3-4+\omega^2)^{10}$$
$$= (-1+\omega^2)^{10}$$
$$= \left(-1-\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)^{10}$$
$$= \left(\frac{3}{2}+i\frac{\sqrt{3}}{2}\right)^{10}$$
$$= (i\sqrt{3})^{10}\left(\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)^{10}$$
$$= -3^5\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)^{10}$$
$$= -3^5\omega^{10}$$

78. Let $S = \frac{1}{2}+\frac{3}{8}+\frac{9}{32}+\frac{27}{128}+\ldots$

$$= \frac{1}{2}\left(1+\frac{3}{4}+\left(\frac{3}{4}\right)^2+\left(\frac{3}{4}\right)^3+\ldots\right)$$
$$= \frac{1}{2}\times\frac{1}{1-\frac{3}{4}}$$
$$= \frac{1}{2}\times\frac{4}{4-3} = 2$$

Thus, $\omega+\omega^s = \omega+\omega^2 = -1$

79. We have,

$z^2+2z^2+2z+1 = 0$

$\Rightarrow$ $(z^3+1)+2z(z+1) = 0$

$\Rightarrow$ $(z+1)(z^2-z+1)+2z(z+1) = 0$

$\Rightarrow$ $(z+1)(z^2-z+1+2z) = 0$

$\Rightarrow$ $(z+1)(z^2+z+1) = 0$

$\Rightarrow$ $z = -1, -\omega, -\omega^2$

When $z = -1$, then

$z^{2013}+z^{2014}+z^{2015}$

$= -1+1-1 = -1 \neq 0$

When $z = \omega$, then

$z^{2013}+z^{2014}+z^{2015}$

$= \omega^{2013}+\omega^{2014}+\omega^{2015}$

$= 1+\omega+\omega^2 = 0$

When $z = \omega^2$, then

$z^{2013}+z^{2014}+z^{2015}$

$= 1+\omega^2+\omega = 0$

Hence, the common roots are ω, ω^2.

80. Let $p = -2013$.

Then $\sqrt[3]{p} = \sqrt[3]{p}, \sqrt[3]{p}\,\omega, \sqrt[3]{\pi}\,\omega^2$

So, $\alpha = \sqrt[3]{p}, \beta = \sqrt[3]{p}\,\omega, \gamma = \sqrt[3]{p}\,\omega^2$

Now, $\frac{x\alpha+y\beta+z\gamma}{x\beta+y\gamma+z\alpha}$

$$= \frac{x \cdot \sqrt[3]{p} + y \cdot \sqrt[3]{p} \cdot \omega + z \cdot \sqrt[3]{p} \cdot \omega^2}{x \cdot \sqrt[3]{p} \cdot \omega + y \cdot \sqrt[3]{p} \cdot \omega^2 + z \cdot \sqrt[3]{p}}$$

$$= \frac{x + y\omega + z\omega^2}{x\omega + y\omega^2 + z}$$

$$= \frac{x + y\omega + z\omega^2}{\omega(x + y\omega + z\omega^2)}$$

$$= \frac{1}{\omega}$$

$$= \omega^2$$

81. We have,

$$\frac{2+3\omega+4\omega^2}{4+3\omega^2+2\omega} = \frac{2+3\omega+4\omega^2}{\omega(2+3\omega+4\omega^2)}$$

$$= \frac{1}{\omega}$$

$$= \omega^2$$

82. We have,

$$\frac{5+6\omega+7\omega^2}{7+6\omega^2+5\omega} + \frac{5+6\omega+7\omega^2}{6+5\omega+7\omega^2}$$

$$= \frac{5+6\omega+7\omega^2}{\omega(5+6\omega+7\omega^2)} + \frac{5+6\omega+7\omega^2}{\omega^2(5+6\omega+7\omega^2)}$$

$$= \frac{1}{\omega} + \frac{1}{\omega^2} = \omega^2 + \omega$$

$$= -1$$

83. We have,

$$4 + 5\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)^{365}$$

$$= 4 + 5\omega^{334} + 3\omega^{365}$$

$$= 4 + 5\omega + 3\omega^2$$

$$= 3(1 + \omega + \omega^2) + 1 + 2\omega$$

$$= 0 + 1 + 2\omega$$

$$= 1 + 2\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$$

$$= 1 - 1 + i\sqrt{3}$$

$$= i\sqrt{3}$$

84. We have

$x^6 - 9x^3 + 8 = 0$

$\Rightarrow$ $(x^3 - 1)(x^3 - 8) = 0$

$\Rightarrow$ $(x^3 - 1) = 0, (x^3 - 8) = 0$

$\Rightarrow$ $x^3 = 1$ and $x^3 = 8$

$\Rightarrow$ $x = 1, \omega, \omega^2$ and $x = 2, 2\omega, 2\omega^2$

$\Rightarrow$ $x = 1, 2, \omega, 2\omega, \omega^2, 2\omega^2$

85. We have,

$x^2 - 2x\cos\theta + 1 = 0$

$$\Rightarrow \quad x = \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2}$$

$$\Rightarrow \quad x = \frac{2\cos\theta \pm 2i\sin\theta}{2} = (\cos\theta \pm i\sin\theta)$$

When $x = \cos\theta + i\sin\theta$, then

$\Rightarrow$ $x^n = (\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$

$$\Rightarrow \quad \frac{1}{x} = \cos\theta - i\sin\theta$$

$$\Rightarrow \quad \frac{1}{x^n} = (\cos\theta - i\sin\theta)^n = \cos(n\theta) - i\sin(n\theta)$$

Thus, $x^n + \frac{1}{x^n} = 2\cos(n\theta)$

Similarly, we can easily prove that,
when $x = \cos\theta - i\sin\theta$, then

$$x^n + \frac{1}{x^n} = 2\cos(n\theta)$$

86. We have,

$$z_r = \cos\left(\frac{2r\pi}{5}\right) + i\sin\left(\frac{2r\pi}{5}\right)$$

$$= e^{i\left(\frac{2r\pi}{5}\right)}$$

Now,

$$z_1 z_2 z_3 z_4 z_5 = e^{i\frac{2\pi}{5}(1+2+3+4+5)}$$

$$= e^{i\frac{2\pi}{5}\times 15}$$

$$= e^{16\pi}$$

$$= \cos(6\pi) + i\sin(6\pi)$$

$$= 1 + i.0$$

$$= 1$$

87. Let $x = \cos\alpha + i\sin\alpha$,
$y = \cos\alpha + i\sin\beta$
and $z = \cos\gamma + i\sin\gamma$

Now, $x + y + z$

$= (\cos\alpha + i\sin\alpha) + (\cos\beta + i\sin\beta) + (\cos\gamma + i\sin\gamma)$

$= (\cos\alpha + \cos\beta + \cos\gamma) + i(\sin\alpha + \sin\beta + \sin\gamma)$

$= 0 + i.0$

$= 0$

$\Rightarrow$ $x^3 + y^3 + z^3 = 3xyz$

$\Rightarrow$ $(\cos\alpha + i\sin\alpha)^3 + (\cos\beta + i\sin\beta)^3 + (\cos\gamma + i\sin\gamma)^3$
$= 3(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta)(\cos\gamma + i\sin\gamma)$

$\Rightarrow$ $(\cos 3\alpha + i\sin 3\alpha) + (\cos 3\beta + i\sin 3\beta) + (\cos 3\gamma + i\sin 3\gamma)$
$= 3(\cos(\alpha + \beta + \gamma) + i\sin(\alpha + \beta + \gamma))$

$\Rightarrow$ $(\cos 3\alpha + \cos 3\beta + \cos 3\gamma) + i(\sin 3\alpha + \sin 3\beta + \sin 3\gamma)$
$= 3\cos(\alpha + \beta + \gamma) + i\,3\sin(\alpha + \beta + \gamma)$

Comparing the real and imaginary parts, we get

$\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3\cos(\alpha + \beta + \gamma)$

and $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3\sin(\alpha + \beta + \gamma)$

Hence, the result.

88. Let $x = \cos\alpha + i\sin\alpha$,
$y = \cos\beta + i\sin\beta$
and $z = \cos\gamma + i\sin\gamma$

Now, $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$
$= \cos\alpha + i\sin\alpha + \cos\beta + i\sin\beta + \cos\gamma + i\sin\gamma$
$= (\cos\alpha + \cos\beta + \cos\gamma) - i(\sin\alpha + \sin\beta + \sin\gamma)$
$= 0 - i.0$
$= 0$

$\Rightarrow \frac{1}{x}+\frac{1}{y}+\frac{1}{z} = 0$

$\Rightarrow \frac{yz+xz+xy}{xyz} = 0$

$\Rightarrow xy + yz + zx = 0$

Also, $(x+y+z)^2 = (x^2+y^2+z^2) + 2(xy+yz+zx)$
$= x^2+y^2+z^2$

$\Rightarrow x^2+y^2+z^2 = 0 \quad (\because x+y+z=0)$

$\Rightarrow (\cos\alpha + i\sin\alpha)^2 + (\cos\beta + i\sin\beta)^2 + (\cos\gamma + i\sin\gamma)^2 = 0$

$\Rightarrow (\cos 2\alpha + i\sin 2\alpha) + (\cos 2\beta + i\sin 2\beta) + (\cos 2\gamma + i\sin 2\gamma) = 0$

$\Rightarrow (\cos 2\alpha + \cos 2\beta + \cos 2\gamma) + (\sin 2\alpha + \sin 2\beta + \sin 2\gamma) = 0 + i.0$

Comparing the real and imaginary parts, we get
$(\cos 2\alpha + \cos 2\beta + \cos 2\gamma) = 0$
and $(\sin 2\alpha + \sin 2\beta + \sin 2\gamma) = 0$

Thus, $(\cos 2\alpha + \cos 2\beta + \cos 2\gamma) = 0$

$\Rightarrow (2\cos^2\alpha - 1 + 2\cos^2\beta - 1 + 2\cos^2\gamma - 1) = 0$

$\Rightarrow (2(\cos^2\alpha + \cos^2\beta + \cos^2\gamma) - 3) = 0$

$\Rightarrow (\cos^2\alpha + \cos^2\beta + \cos^2\gamma) = \frac{3}{2}$

and

$(1-\sin^2\alpha) + (1-\sin^2\beta) + (1-\sin^2\gamma) = \frac{3}{2}$

$\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 3 - \frac{3}{2} = \frac{3}{2}$

Hence, the result.

89. Given $z_r = \cos\left(\frac{\pi}{2^r}\right) + i\sin\left(\frac{\pi}{2^r}\right) = e^{i\left(\frac{\pi}{2^r}\right)}$

Now, $z_1.z_2.z_3 \ldots$ to ∞

$= e^{i\left(\frac{\pi}{2}\right)} \cdot e^{i\left(\frac{\pi}{2^2}\right)} \cdot e^{i\left(\frac{\pi}{2^3}\right)} \ldots$ to ∞

$= e^{i\left(\frac{\pi}{2}\right)\left(1+\frac{1}{2}+\frac{1}{2^2}+\ldots\text{to}\infty\right)}$

$= e^{i\left(\frac{\pi}{2}\right)\left(\frac{1}{1-\frac{1}{2}}\right)}$

$= e^{i\left(\frac{\pi}{2}\right)\times 2i}$

$= e^{i\pi}$

$= \cos(\pi) + i\sin(\pi)$

$= -1$

90. We have,

$(1+i) = \sqrt{2}\left(\frac{1}{\sqrt{2}}+\frac{i}{\sqrt{2}}\right)$

$= \sqrt{2}\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right)$

$\Rightarrow (1+i)^8 = (\sqrt{2})^8\left(\cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right)\right)^8$

$= 2^4\left(\cos\left(\frac{\pi}{4}\times 8\right) + i\sin\left(\frac{\pi}{4}\times 8\right)\right)$

$= 24(\cos(2\pi) + i\sin(2\pi))$

Similarly,

$(1-i)^8 = 2^4(\cos(2\pi) - i\sin(2\pi))$

Therefore, $(1+i)^8 + (1-i)^8$
$= 2^4(\cos(2\pi) + i\sin(2\pi))$
$+ 2^4[\cos(2\pi) - i\sin(2\pi)]$
$= 2^4[2\cdot\cos(2\pi)]$
$= 2^5\cdot 1$
$= 32$

91. We have,

$$\sum_{k=1}^{21}\left(\sin\left(\frac{2\pi k}{11}\right) - i\cos\left(\frac{2\pi k}{11}\right)\right)$$

$$= \sum_{k=1}^{21}\left(-i^2\sin\left(\frac{2\pi k}{11}\right) - i\cos\left(\frac{2\pi k}{11}\right)\right)$$

$$= \sum_{k=1}^{21} -i\left(\cos\left(\frac{2\pi k}{11}\right) + i\sin\left(\frac{2\pi k}{11}\right)\right)$$

$$= \sum_{k=1}^{21} -ie^{i\left(\frac{2\pi k}{11}\right)}$$

$$= -i\left(\sum_{k=1}^{21} e^{i\left(\frac{2\pi k}{11}\right)}\right)$$

$$= -i\left(e^{i\left(\frac{2\pi}{11}\right)} + e^{i\left(\frac{4\pi}{11}\right)} + e^{i\left(\frac{6\pi}{11}\right)} + \ldots + e^{e^{i\left(\frac{42\pi}{11}\right)}}\right)$$

$$= -ie^{i\left(\frac{2\pi}{11}\right)}\left(1 + e^{i\left(\frac{2\pi}{11}\right)} + e^{i\left(\frac{4\pi}{11}\right)} + \ldots + e^{i\left(\frac{40\pi}{11}\right)}\right)$$

$$= -ie^{i\left(\frac{2\pi}{11}\right)}\left(\frac{1-e^{i\left(\frac{42\pi}{11}\right)}}{1-e^{i\left(\frac{2\pi}{11}\right)}}\right)$$

$$= -i\left(\frac{e^{i\left(\frac{2\pi}{11}\right)} - e^{i\left(\frac{44\pi}{11}\right)}}{1-e^{i\left(\frac{2\pi}{11}\right)}}\right)$$

$$= -i\left(\frac{e^{i\left(\frac{2\pi}{11}\right)} -}{1-e^{i\left(\frac{2\pi}{11}\right)}}\right)$$

$= -i \times -1$
$= i$

92. We have

$x^3 - 1 = 0$

$\Rightarrow \quad x^3 = 1 = \cos(2r\pi) + i\sin(2r\pi)$

$$\Rightarrow \quad x = \cos\left(\frac{2r\pi}{3}\right) + i\sin\left(\frac{2r\pi}{3}\right), r = 0, 1, 2$$

When $r = 0$, $x = 1$

When $r = 1$, $x = \cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right) = e^{i\frac{2\pi}{3}}$

When $r = 2$, $x = \cos\left(\frac{4\pi}{3}\right) + i\sin\left(\frac{4\pi}{3}\right) = e^{i\frac{4\pi}{3}}$

Hence, the solutions are $\left\{1, e^{i\frac{2\pi}{3}}, e^{i\frac{4\pi}{3}}\right\}$.

93. We have $x^5 - 1 = 0$

$\Rightarrow \quad x^5 = 1 = \cos(2r\pi) + i\sin(2r\pi)$

$$\Rightarrow \quad x = \cos\left(\frac{2r\pi}{5}\right) + i\sin\left(\frac{2r\pi}{5}\right)$$

Where $r = 0, 1, 2, 3, 4$

When $r = 0$, then $x = 1$

When $r = 1$, $x = \cos\left(\frac{2\pi}{5}\right) + i\sin\left(\frac{2\pi}{5}\right) = e^{i\frac{2\pi}{5}}$

When $r = 2$, $x = \cos\left(\frac{4\pi}{5}\right) + i\sin\left(\frac{4\pi}{5}\right) = e^{i\frac{4\pi}{5}}$

When $r = 3$, $x = \cos\left(\frac{6\pi}{5}\right) + i\sin\left(\frac{6\pi}{5}\right)$

$$= \cos\left(2\pi - \frac{4\pi}{5}\right) + i\sin\left(2\pi - \frac{4\pi}{5}\right)$$

$$= \cos\left(\frac{4\pi}{5}\right) - i\sin\left(\frac{4\pi}{5}\right) = e^{-i\frac{4\pi}{5}}$$

When $r = 4$, $x = \cos\left(\frac{8\pi}{5}\right) + i\sin\left(\frac{8\pi}{5}\right)$

$$= \cos\left(2\pi - \frac{2\pi}{5}\right) + i\sin\left(2\pi - \frac{2\pi}{5}\right)$$

$$= \cos\left(\frac{2\pi}{5}\right) - i\sin\left(\frac{2\pi}{5}\right) = e^{-i\frac{2\pi}{5}}$$

Hence, the solutions are $\left\{1, e^{\pm i\frac{2\pi}{5}}, e^{\pm i\frac{4\pi}{5}}\right\}$,

i.e. $\left\{e^{\pm i\frac{2r\pi}{5}}\right\}$, where $r = 0, 1, 2$.

94. We have,

$x^7 - 1 = 0$

$$\Rightarrow \quad x = \cos\left(\frac{2r\pi}{7}\right) + i\sin\left(\frac{2r\pi}{7}\right)$$

where $r = 0, 1, 2, 3, 4, 5, 6$

$\Rightarrow \quad x = e^{i\frac{2r\pi}{7}}, r = 0, 1, 2, 3, 4, 5, 6$

Hence, the solution set is

$\left\{e^{\pm i\frac{2r\pi}{7}}\right\}$, where $r = 0, 1, 2, 3$

95. We have,

$x^3 + 1 = 0$

$\Rightarrow \quad x^3 = -1 = \cos(2r + 1)\pi + i\sin(2r + 1)\pi$

where $r = 0, 1, 2$

$$\Rightarrow \quad x = \cos\left(\frac{2r+1}{3}\right)\pi + i\sin\left(\frac{2r+1}{3}\right)\pi$$

When $r = 0$, $x = \cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right)$

$$p = -\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = -\omega^2$$

When $r = 1$, $x = -1$

When $r = 2$, $x = \cos\left(\frac{5\pi}{3}\right) + i\sin\left(\frac{5\pi}{3}\right)$

$$= \cos\left(2\pi - \frac{\pi}{3}\right) + i\sin\left(2\pi - \frac{\pi}{3}\right)$$

$$= \cos\left(\frac{\pi}{3}\right) - i\sin\left(\frac{\pi}{3}\right)$$

$$= \frac{1}{2} - i\frac{\sqrt{3}}{2} = -\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

$$= -\omega$$

Hence, the solutions are $\{-1, -\omega, -\omega^2\}$.

96. We have,

$x^5 + 1 = 0$

$$\Rightarrow \quad x = \cos\left(\frac{2r+1}{5}\right)\pi + i\sin\left(\frac{2r+1}{5}\right)\pi$$

Where $r = 0, 1, 2, 3, 4$.

When $r = 0$, $x = \cos\left(\frac{\pi}{5}\right) + i\sin\left(\frac{\pi}{5}\right) = e^{i\frac{\pi}{5}}$

When $r = 1$, $x = \cos\left(\frac{3\pi}{5}\right) + i\sin\left(\frac{3\pi}{5}\right) = e^{i\frac{3\pi}{4}}$

When $r = 2$, $x = -1$

When $r = 3$, $x = \cos\left(\frac{7\pi}{5}\right) + i\sin\left(\frac{7\pi}{5}\right)$

$$= \cos\left(2\pi - \frac{3\pi}{5}\right) + i\sin\left(2\pi - \frac{3\pi}{5}\right)$$

$$= \cos\left(\frac{3\pi}{5}\right) - i\sin\left(\frac{3\pi}{5}\right) = e^{-i\frac{3\pi}{5}}$$

When $r = 4$, $x = \cos\left(\frac{9\pi}{5}\right) + i\sin\left(\frac{9\pi}{5}\right)$

$$x = \cos\left(2\pi - \frac{\pi}{5}\right) + i\sin\left(2\pi - \frac{\pi}{5}\right)$$

$$= \cos\left(\frac{\pi}{5}\right) - i\sin\left(\frac{\pi}{5}\right) = e^{-i\frac{\pi}{5}}$$

Hence, the solutions are

$$\left\{-1, e^{\pm i\frac{\pi}{5}}, e^{\pm i\frac{3\pi}{5}}\right\}$$

i.e. $\left\{e^{\pm i\left(\frac{2r+1}{5}\right)\pi}\right\}, r = 0, 1, 2$

97. We have,

$$x^7+1=0$$

$$x=\cos\left(\frac{2r+1}{7}\right)\pi+i\sin\left(\frac{2r+1}{7}\right)\pi$$

where $r = 0, 1, 2, 3, 4, 5, 6.$

$$\left\{e^{\pm i\left(\frac{2r+1}{7}\right)\pi}\right\}, r=0,1,2,3$$

Hence, the solutions set is

$$\left\{-1, e^{\pm i\left(\frac{\pi}{7}\right)}, e^{\pm i\left(\frac{3\pi}{7}\right)}, e^{\pm i\left(\frac{5\pi}{7}\right)}\right\}$$

98. We have,

$$x^{10}-1=0$$

$\Rightarrow$ $(x^5+1)(x^5-1)=0$

$\Rightarrow$ $x^5-1=0$ and $x^5+1=0$

When $x^5-1=0$,

$$\Rightarrow \quad x=\cos\left(\frac{2r\pi}{5}\right)+i\sin\left(\frac{2r\pi}{5}\right)=e^{i\frac{2r\pi}{5}}$$

Where $r = 0, 1, 2, 3, 4$

$$\left\{e^{\pm i\frac{2r\pi}{5}}\right\} \text{ where } r=0,1,2$$

When $x^5+1=0$

$$\Rightarrow \quad x=\cos\left(\frac{2r+1}{5}\right)\pi+i\sin\left(\frac{2r+1}{5}\right)\pi$$

$$\Rightarrow \quad x=e^{i\left(\frac{2r+1}{5}\right)\pi}, r=0,1,2,3,4$$

Thus $\left\{e^{\pm i\frac{(2r+1)\pi}{5}}\right\}$ where $r = 0, 1, 2$

Hence, the solutions are

$$\left\{e^{\pm i\left(\frac{2r\pi}{5}\right)}, e^{\pm i\left(\frac{(2r+1)\pi}{5}\right)}\right\}, r=0,1,2$$

99. We have,

$$x^{10}+x^5+1=0$$

$\Rightarrow$ $(x^5)^2+(x^5)+1=0$

$\Rightarrow$ $(x^5)=\omega, \omega^2$

When $(x^5)=\omega$

$$\Rightarrow \quad x^5=-\frac{1}{2}+i\frac{\sqrt{3}}{2}$$

$$x^5=\cos\left(2r\pi+\frac{2\pi}{3}\right)+i\sin\left(2r\pi+\frac{2\pi}{3}\right)$$

$$\Rightarrow \quad x=\cos\left(\frac{2r\pi+\frac{2\pi}{3}}{5}\right)+i\sin\left(\frac{2r\pi+\frac{2\pi}{3}}{5}\right)$$

$$=\cos\left(\frac{6r\pi+2\pi}{15}\right)+i\sin\left(\frac{6r\pi+2\pi}{15}\right)$$

where $r = 0, 1, 2, 3, 4$

$$\Rightarrow \quad x=e^{i\left(\frac{6r\pi+2\pi}{15}\right)}, r=0,1,2,3,4$$

Similarly, $(x^5)=\omega^2$ will provide us

$$x=\cos\left(\frac{(2r+1)\pi+\frac{4\pi}{3}}{5}\right)+i\sin\left(\frac{(2r+1)\pi+\frac{4\pi}{3}}{5}\right)$$

$$\Rightarrow \quad =\cos\left(\frac{3(2r+1)\pi+4\pi}{15}\right)+i\sin\left(\frac{3(2r+1)\pi+4\pi}{15}\right)$$

$$=\cos\left(\frac{6r\pi+7\pi}{15}\right)+i\sin\left(\frac{6r\pi+7\pi}{15}\right)$$

$$=e^{i\left(\frac{6r\pi+7\pi}{15}\right)}, r=0,1,2,3,4$$

Hence, the solutions set is

$$\left\{e^{i\left(\frac{6r\pi+2\pi}{15}\right)}, e^{i\left(\frac{6r\pi+7\pi}{15}\right)}\right\} r=0,1,2,3,4$$

100. We have

$$x^{10}-x^9+x^8-x^7+\ldots+x^2-x+1=0$$

$\Rightarrow$ $1-x+x^2-x^3+\ldots-x^{10}=0$

$$\Rightarrow \quad \frac{1-(-x)^{11}}{1-(-x)}=0$$

$\Rightarrow$ $x^{11}=-1$

$$\Rightarrow \quad x=\cos\left(\frac{(2r+1)\pi}{11}\right)+i\sin\left(\frac{(2r+1)\pi}{11}\right)$$

$$=e^{i\left(\frac{(2r+1)\pi}{11}\right)}, r=0,1,2,\ldots,10$$

Hence , the solutions are

$$\left\{e^{\pm i\left(\frac{(2r+1)\pi}{11}\right)}\right\}, r=0,1,2,3,4,5$$

101. We have $z^5+1=0$

Hence, the solutions of z are

$$\left\{-1, e^{\pm i\frac{\pi}{5}}, e^{\pm i\frac{3\pi}{5}}\right\}$$

$= \{-1, \alpha, \bar{\alpha}, \beta, \bar{\beta}\}$, where $\alpha, \beta \in C$

Now, $\alpha+\bar{\alpha}=2\cos\left(\frac{\pi}{5}\right)$, $\alpha\cdot\bar{\alpha}=1$

and $\beta+\bar{\beta}=2\cos\left(\frac{3\pi}{5}\right)$, $\beta\cdot\bar{\beta}=1$

Thus, z^5+1

$=(z+1)(z-\alpha)(z-\bar{\alpha})(z-\beta)(z-\bar{\beta})$

$=(z+1)(z^2-(\alpha+\bar{\alpha})z+\alpha\cdot\bar{\alpha})(z^2-(\beta+\bar{\beta})z+\beta\cdot\bar{\beta})$

$$=(z+1)\left(z^2-2\cos\left(\frac{\pi}{5}\right)z+1\right)\left(z^2-2\cos\left(\frac{3\pi}{5}\right)z+1\right)$$

$$\Rightarrow \frac{z^5+1}{(z+1)}=\left(z^2-2\cos\left(\frac{\pi}{5}\right)z+1\right)\left(z^2-2\cos\left(\frac{3\pi}{5}\right)z+1\right)$$

Put $z=i$, we get,

$$\Rightarrow \frac{i+1}{(i+1)}=\left(-1-2\cos\left(\frac{\pi}{5}\right)i+1\right)\left(-1-2\cos\left(\frac{3\pi}{5}\right)i+1\right)$$

$$\Rightarrow \quad 1=\left(-2\cos\left(\frac{\pi}{5}\right)i\right)\left(-2\cos\left(\frac{3\pi}{5}\right)\right)$$

$$=\left(-2\cos\left(\frac{\pi}{5}\right)i\right)\left(-2\cos\left(\frac{3\pi}{5}\right)i\right)$$

$$\Rightarrow \quad 4\cos\left(\frac{\pi}{5}\right)\cos\left(\frac{3\pi}{5}\right)=-1$$

$$\Rightarrow \quad 4\cos\left(\frac{\pi}{5}\right)\cos\left(\pi-\frac{2\pi}{5}\right)=-1$$

$$\Rightarrow \quad 4\cos\left(\frac{\pi}{5}\right)\cos\left(\frac{2p}{5}\right)=1$$

$$\Rightarrow \quad 4\cos\left(\frac{\pi}{5}\right)\cos\left(\frac{\pi}{2}-\frac{\pi}{10}\right)=1$$

$$\Rightarrow \quad 4\cos\left(\frac{\pi}{5}\right)\sin\left(\frac{\pi}{10}\right)=1$$

Hence, the result.

102. We have,

$$z^7-1=0$$

$$\Rightarrow \quad z=\cos\left(\frac{2r\pi}{7}\right)+i\sin\left(\frac{2r\pi}{7}\right)$$

where $r = 0, 1, 2, 3, 4, 5, 6$

$$\Rightarrow \quad z=e^{i\frac{2r\pi}{7}},\ r=0,1,2,3,4,5,6$$

Hence, the solution set is

$$\left\{e^{\pm i\frac{2r\pi}{7}}\right\},\ \text{where } r=0,1,2,3$$

$$=\left\{1, e^{\pm i\frac{2\pi}{7}}, e^{\pm i\frac{4\pi}{7}}, e^{\pm i\frac{6\pi}{7}}\right\}$$

$$=\{1, \alpha, \bar{\alpha}, \beta, \bar{\beta}, \gamma, \bar{\gamma}\}$$

Thus, $\alpha+\bar{\alpha}=2\cos\left(\frac{2\pi}{7}\right),\ \alpha\cdot\bar{\alpha}=1$

$$\beta+\bar{\beta}=2\cos\left(\frac{4\pi}{7}\right),\ \beta\cdot\bar{\beta}=1$$

and $\gamma+\bar{\gamma}=2\cos\left(\frac{6\pi}{7}\right),\ \gamma\cdot\bar{\gamma}=1$

Thus, z^7-1

$$=(z-1)(z-\alpha)(z-\bar{\alpha})(z-\beta)(z-\bar{\beta})(z-\gamma)(z-\bar{\gamma})$$

$$\Rightarrow \quad \frac{z^7-1}{z-1}=(z^2-(\alpha+\bar{\alpha})z+\alpha\cdot\bar{\alpha})$$

$$\times(z^2-(\beta-\bar{\beta})z+\beta\bar{\beta})(z^2-(\gamma+\bar{\gamma})(z+\gamma\cdot\bar{\gamma})$$

$$=\left(z^2-2\cos\left(\frac{2\pi}{7}\right)z+1\right)$$

$$\times\left(z^2-2\cos\left(\frac{4\pi}{7}\right)z+1\right)\left(z^2-2\cos\left(\frac{6\pi}{7}\right)z+1\right)$$

Put $z = i$, we get

$$\frac{-(i+1)}{i-1}$$

$$=\left(-2\cos\left(\frac{2\pi}{7}\right)i\right)\left(-2\cos\left(\frac{4\pi}{7}\right)i\right)\left(-2\cos\left(\frac{6\pi}{7}\right)i\right)$$

$$=\left(-8\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\frac{6\pi}{7}\right)i\right)$$

$$\Rightarrow \quad \frac{(i+1)}{i-1}=\left(8\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\frac{6\pi}{7}\right)i\right)$$

$$\Rightarrow \quad \frac{(i+1)^2}{(i)^2-(1)^2}=8\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\frac{6\pi}{7}\right)i$$

$$\Rightarrow \quad \frac{2i}{-2}=\left(8\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\frac{6\pi}{7}\right)i\right)$$

$$\Rightarrow \quad 8\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\frac{6\pi}{7}\right)=-1$$

$$\Rightarrow \quad 8\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\pi-\frac{\pi}{7}\right)=-1$$

$$\Rightarrow \quad 8\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)\cos\left(\frac{\pi}{7}\right)=1$$

$$\Rightarrow \quad \cos\left(\frac{\pi}{7}\right)\cos\left(\frac{2\pi}{7}\right)\cos\left(\frac{4\pi}{7}\right)=\frac{1}{8}$$

103. We have

$$(1-i)^x=2^x$$

$$\Rightarrow \quad |(1-i)|^x=|2^x|$$

$$\Rightarrow \quad (\sqrt{2})^x=2^x$$

$$\Rightarrow \quad 2^{\frac{x}{2}}=2^x$$

$$\Rightarrow \quad 2^{x-\frac{x}{2}}=1$$

$$\Rightarrow \quad 2^{\frac{x}{2}}=1=2^0$$

$$\Rightarrow \quad \frac{x}{2}=0$$

$$\Rightarrow \quad x=0$$

Hence, the integral solution is $\{0\}$.

104. We have,

$$z=\left(\frac{\sqrt{3}-i}{2}\right)$$

$$=i\left(-\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)=i\omega^2$$

Now,

$$z^{101}+z^{103}=i^{101}\omega^{202}+i^{103}z^{206}$$

$$=i\omega+i^3\omega^2$$

$$=i\omega-i\omega^2$$

$$=i(\omega-\omega^2)$$

$$=i\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}-\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)$$

$$=i(i\sqrt{3})$$

$$=-\sqrt{3}$$

105. We have,

$$\bar{z}z^3 + z\overline{z^3} = 350$$

$\Rightarrow$ $|z|^2 z^2 + |z|^2 \overline{z^2} = 350$

$\Rightarrow$ $|z|^2 (z^2 + \overline{z^2}) = 350$

$\Rightarrow$ $2(x^2 + y^2)(x^2 - y^2) = 350$

$\Rightarrow$ $(x^2 + y^2)(x^2 - y^2) = 175$

$\Rightarrow$ $(x^2 + y^2)(x^2 - y^2) = 25 \times 7$

$\Rightarrow$ $(x^2 + y^2) = 25$ and $(x^2 - y^2) = 7$

$\Rightarrow$ $x = 4$ and $y = 3$

Thus, the area of the rectangle $= 2x \times 2y$

$= 48$ sq. u.

106. We have,

$$\sum_{m=1}^{15} \text{Im}(z^{2m-1})$$

$= \text{Im}(z + z^3 + z^5 + \ldots + z^{29})$

$= \sin\theta + \sin 3\theta + \sin 5\theta + \ldots + \sin 29\theta$

$= \dfrac{\sin(15\theta)}{\sin(\theta)} \times \sin(\theta + 14\theta)$

$= \dfrac{\sin^2(15\theta)}{\sin(\theta)}$

$= \dfrac{\sin^2(30°)}{\sin(2^{\circ})}$

$= \dfrac{1}{4\sin(2^{\circ})}$

107. We have,

$z^{p+q} - z^p - z^q + 1 = 0$

$\Rightarrow$ $z^p(z^q - 1) - 1(z^q - 1) = 0$

$\Rightarrow$ $(z^p - 1)(z^q - 1) = 0$

$\Rightarrow$ either $(z^p - 1) = 0$ or $(z^q - 1) = 0$

$\Rightarrow$ either $(z - 1)(1 + z + z^2 + \ldots + z^{p-1}) = 0$

or $(z - 1)(1 + z + z^2 + \ldots + z^{q-1}) = 0$

$\Rightarrow$ either $(1 + z + z^2 + \ldots + z^{p-1}) = 0$

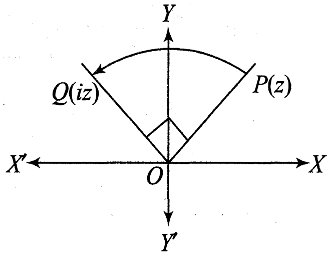
or $(1 + z + z^2 + \ldots + z^{q-1}) = 0$

$\Rightarrow$ either $(1 + \alpha + \alpha^2 + \ldots + \alpha^{p-1}) = 0$

or $(1 + \alpha + \alpha^2 + \ldots + \alpha^{q-1}) = 0$

108. Let $z = 3 + 4i$.

If it is rotated through an angle of 90°, then it is converted to $ze^{i\frac{\pi}{2}} = iz$



Thus, $iz = i(3 + 4i) = 3i - 4 = -4 + 3i = (-4, 3)$.

Hence, the new position of P is $(-4, 3)$.

109. Let $z = 3 + 4i$.

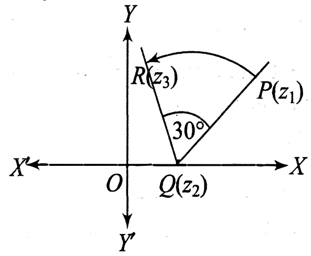
If it is rotated through an angle of 180°, then it is converted to $ze^{i\pi} = -z$

Thus, $-z = -(3 + 4i) = (-3 - 4i) = (-3, -4)$.

Hence, the new position of Q is $(-3, -4)$

110. Let the new position of P be R.

Also, let P, Q, R represent the complex numbers z_1, z_2, z_3 respectively.



By Coni method,

$$\left(\frac{z_3 - z_2}{z_1 - z_2}\right) = \left|\frac{z_3 - z_2}{z_1 - z_2}\right| \times e^{i\frac{\pi}{6}}$$

$\Rightarrow$ $\left(\dfrac{z_3 - z_2}{z_1 - z_2}\right) = \dfrac{|z_3 - z_2|}{|z_1 - z_2|} \times e^{i\frac{\pi}{6}}$

$\Rightarrow$ $\left(\dfrac{z_3 - z_2}{z_1 - z_2}\right) = e^{i\frac{\pi}{6}} = \cos\left(\dfrac{\pi}{6}\right) + i\sin\left(\dfrac{\pi}{6}\right)$

$\Rightarrow$ $\left(\dfrac{z_3 - z_2}{z_1 - z_2}\right) = \dfrac{\sqrt{3}}{2} + \dfrac{i}{2}$

$\Rightarrow$ $|z_3 - z_2| = 2(1 + 2i) \times \dfrac{1}{2}(\sqrt{3} + i)$

$\Rightarrow$ $= (1 + 2i)(\sqrt{3} + i)$

$\Rightarrow$ $= (\sqrt{3} + i + 2\sqrt{3}i - 2)$

$\Rightarrow$ $z_3 = z_2 + (\sqrt{3} - 2) + i(1 + 2\sqrt{3})$

$= 1 + (\sqrt{3} - 2) + i(1 + 2\sqrt{3})$

$= (\sqrt{3} - 1) + i(1 + 2\sqrt{3})$

$= ((\sqrt{3} - 1), (1 + 2\sqrt{3}))$

Hence, the new position of P is

$[(\sqrt{3} - 1), (1 + 2\sqrt{3})]$

111. Let $z_1 = \sqrt{3} + i$ and $z_2 = -1 + i\sqrt{3}$

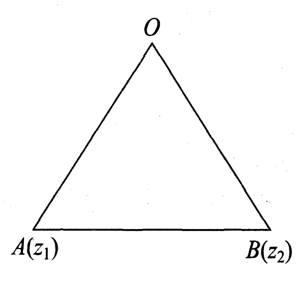
Now, $z_2 = -1 + i\sqrt{3} = i(i + \sqrt{3}) = iz_1$

Again, $i = 0 + i \cdot 1 = \cos\left(\dfrac{\pi}{2}\right) + i\sin\left(\dfrac{\pi}{2}\right) = e^{i\frac{\pi}{2}}$

Thus, $\theta = \dfrac{\pi}{2}$

112. As we know that if z_1, z_2, z_3 are the vertices of an equilateral triangle, then

$$z_1^2 + z_2^2 + z_3^2 - z_1z_2 - z_2z_3 - z_3z_1 = 0$$



Putting $z_3 = 0$, we get,

$$z_1^2 + z_2^2 = z_1 z_2$$

Hence, the result.

113. Let z_1 and z_2 are the roots of $z^2 + az + b = 0$.

Then $z_1 + z_2 = -a$, $z_1 z_2 = b$

As we know that, if $0, z_1, z_2$ represent the vertices of an equilateral triangle, then

$$z_1^2 + z_2^2 = z_1 z_2$$

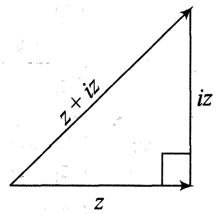
$\Rightarrow \quad (z_1 + z_2)^2 - 2z_1 z_2 = z_1 z_2$

$\Rightarrow \quad (z_1 + z_2)^2 = 3z_1 z_2$

$\Rightarrow \quad (-a)^2 = 3b$

$\Rightarrow \quad a^2 = 3b$

114. Clearly, it is a right-angled triangle.



Thus,

Area of the given triangle

$$= \frac{1}{2} \times |z| \times |iz|$$

$$= \frac{1}{2} \times |z|^2 \times |i|$$

$$= \frac{1}{2} \times |z|^2$$

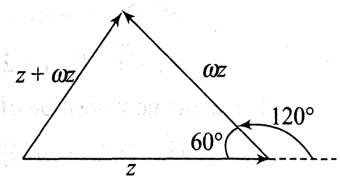
It is given that, $\frac{1}{2} \times |z|^2 = 50$

$\Rightarrow \quad |z|^2 = 100$

$\Rightarrow \quad |z| = 10$

Hence, the value of $|z|$ is 10.

115.



We have $|\omega z| = |\omega||z|$ $= |z|$

Also, $|z + \omega z|$

$= |(1 + \omega)||z|$

$= |-\omega^2||z| = |z|$

Thus, it will form an equilateral triangle.

Hence,

$$\text{Area} = \frac{\sqrt{3}}{4} \times (\text{side})^2$$

$\Rightarrow \quad \frac{\sqrt{3}}{4} \times (\text{side})^2 = 16\sqrt{3}$

$\Rightarrow \quad \frac{\sqrt{3}}{4} \times |z|^2 = 16\sqrt{3}$

$\Rightarrow \quad |z|^2 = 64$

$\Rightarrow \quad |z| = 8$

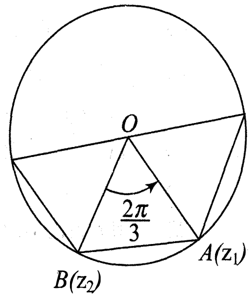
Thus, the value of

$$(|z|^2 + |z| + 2) = 64 + 8 + 2 = 74 \text{ sq. u.}$$

116. Let $z = (1)^{1/n} = [\cos(2r\pi) + i\sin(2r\pi)]^{1/n}$

$$= \left(\cos\left(\frac{2r\pi}{n}\right) + i\sin\left(\frac{2r\pi}{n}\right)\right)$$

where $r = 0, 1, 2, 3, \ldots, (n-1)$



Let

$$z_1 = 1 \text{ and } z_2 = e^{i\frac{2k\pi}{n}}.$$

It is given that,

$$(z_2 - 0) = (z_1 - 0)e^{i\frac{\pi}{2}}$$

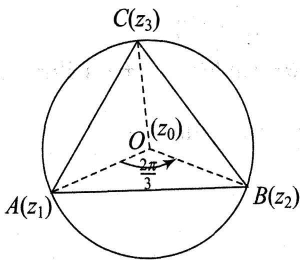
$\Rightarrow \quad e^{i\frac{2k\pi}{n}} = e^{i\frac{\pi}{2}}$

$\Rightarrow \quad \frac{2k\pi}{n} = \frac{\pi}{2}$

$\Rightarrow \quad n = 4k$

Hence, the result.

117.



We have,

$$\left(\frac{z_2 - z_0}{z_1 - z_0}\right) = \left|\frac{z_2 - z_0}{z_1 - z_0}\right| \times e^{i\frac{2\pi}{3}}$$

$\Rightarrow \quad \left(\frac{z_2 - z_0}{z_1 - z_0}\right) = \frac{|z_2 - z_0|}{|z_1 - z_0|} \times e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad \left(\frac{z_2}{z_1}\right) = e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad z_2 = z_1 e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad z_2 = (1 + i\sqrt{3})\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$

$\Rightarrow \quad z_2 = \frac{1}{2}(1 + i\sqrt{3})(-1 + i\sqrt{3})$

$\Rightarrow \quad z_2 = -\frac{1}{2}(1 + 3) = -2$

Also, $\left(\frac{z_3 - z_0}{z_2 - z_0}\right) = \left|\frac{z_3 - z_0}{z_2 - z_0}\right| \times e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad \left(\frac{z_3 - z_0}{z_2 - z_0}\right) = \frac{|z_3 - z_0|}{|z_2 - z_0|} \times e^{i\frac{2\pi}{3}}$

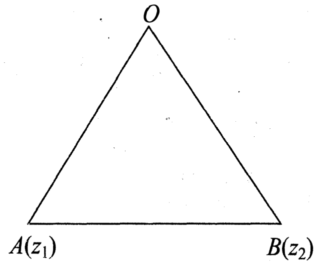
$\Rightarrow \quad \left(\frac{z_3}{z_2}\right) = e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad z_3 = z_2 \times e^{i\frac{2\pi}{3}} = -2\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right)$

$\Rightarrow \quad z_3 = -2\left(-\frac{1}{2} + \frac{i\sqrt{3}}{2}\right) = (1 - i\sqrt{3})$

118. As we know that, if $z_1, z_2, 0$ represent the vertices of an equilateral triangle, then

$$z_1^2 + z_2^2 = z_1 z_2$$



$\Rightarrow \quad (a+i)^2 + (1+ib)^2 = (a+i)(1+ib)$

$\Rightarrow \quad (a^2 - 1 + 2ia) + (1 - b^2 + 2ib) = (a-b) + i(1+ab)$

$\Rightarrow \quad (a^2 - b^2 + 2i(a-b)) = (a-b) + i(1+ab)$

Comparing the real and imaginary parts, we get,

$(a^2 - b^2) = (a-b)$ and $2(a+b) = (1+ab)$

$\Rightarrow \quad a = b$ and $2(a+b) = 1 + ab$

$\Rightarrow \quad a^2 - 4a + 1 = 0$

$\Rightarrow \quad (a-2)^2 = (\sqrt{3})^2$

$\Rightarrow \quad (a-2) = \pm\sqrt{3}$

$\Rightarrow \quad a = 2 \pm \sqrt{3}$

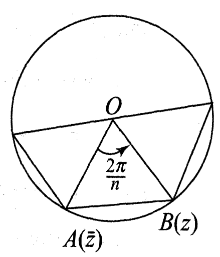
$\Rightarrow \quad a = 2 - \sqrt{3} = b$

119. Let $z = \cos\theta + i\sin\theta$

Then $\bar{z} = \cos\theta - i\sin\theta$

We have $\left(\dfrac{z - z_0}{\bar{z} - z_0}\right) = \left|\dfrac{z - z_0}{\bar{z} - z_0}\right| \times e^{i\frac{2\pi}{n}}$

$\Rightarrow \quad \left(\dfrac{z - z_0}{\bar{z} - z_0}\right) = \dfrac{|z - z_0|}{|\bar{z} - z_0|} \times e^{i\frac{2\pi}{n}}$



$\Rightarrow \quad \left(\dfrac{z}{\bar{z}}\right) = e^{i\frac{2\pi}{n}}$

$\Rightarrow \quad e^{i2\theta} = e^{i\frac{2\pi}{n}}$

$\Rightarrow \quad 2\theta = \dfrac{2\pi}{n}$

$\Rightarrow \quad \theta = \dfrac{\pi}{n}$ ...(i)

Also, $\dfrac{\operatorname{Im}(z)}{\operatorname{Re}(z)} = \sqrt{2} - 1$

$\Rightarrow \quad \tan\theta = \sqrt{2} - 1 = \tan\left(\dfrac{\pi}{8}\right)$

$\Rightarrow \quad \theta = \dfrac{\pi}{8}$

$\Rightarrow \quad \dfrac{\pi}{n} = \dfrac{\pi}{8}$ [from Eq. (i)]

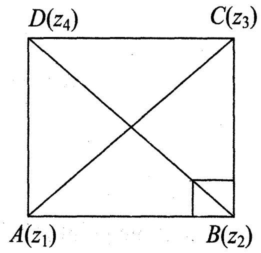
$\Rightarrow \quad n = 8$

Hence, the value of n is 8.

120. Let $A(z_1)$, $B(z_2)$, $C(z_3)$ and $D(z_4)$ represent the vertices of a square $ABCD$ respectively.

We have $\left(\dfrac{z_3 - z_2}{z_1 - z_2}\right) = \left|\dfrac{z_3 - z_2}{z_1 - z_2}\right| \times e^{-i\frac{\pi}{2}}$

$\Rightarrow \quad \left(\dfrac{z_3 - z_2}{z_1 - z_2}\right) = e^{-i\frac{\pi}{2}}$



$\Rightarrow \quad \left(\dfrac{z_3 - z_2}{z_1 - z_2}\right) = -i$

$\Rightarrow \quad (z_3 - z_2) = -i(z_1 - z_2)$

$\Rightarrow \quad (1+i)z_2 = z_3 + iz_1$

$= 3 - 2i - 2 + 2i = 0$

$\Rightarrow \quad z_2 = \dfrac{0}{(1+i)} = 0 = (0, 0)$

The diagonals AC and BD bisect each other and they intersect at O, whose coordinates are $\left(\dfrac{5}{2}, \dfrac{1}{2}\right)$.

Let $z_4 = \alpha + i\beta = (\alpha, \beta)$

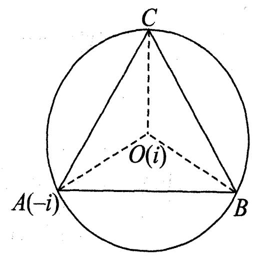
Now, $\dfrac{\alpha + 0}{2} = \dfrac{5}{2} \quad \Rightarrow \quad \alpha = 5$

and $\dfrac{\beta + 0}{2} = \dfrac{1}{2} \quad \Rightarrow \quad \beta = 1$

Thus, the coordinates of

$B = (0, 0)$ and $D = (5, 1)$

121. Let $A(z_1)$, $B(z_2)$, $C(z_3)$ represent the vertices of an equilateral triangle and its centre is $z_0 = i$.



We have,

$\left(\dfrac{z_2 - z_0}{z_1 - z_0}\right) = \left|\dfrac{z_2 - z_0}{z_1 - z_0}\right| \times e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad \left(\dfrac{z_2 - z_0}{z_1 - z_0}\right) = \dfrac{|z_2 - z_0|}{|z_1 - z_0|} \times e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad \left(\dfrac{z_2 - z_0}{z_1 - z_0}\right) = e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad (z_2 - z_0) = (z_1 - z_0) \times e^{i\frac{2\pi}{3}}$

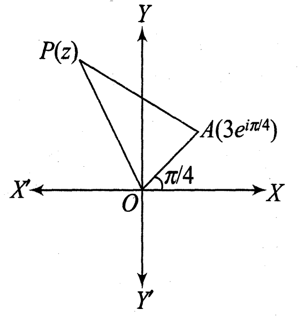
$\Rightarrow \quad z_2 = z_0 + (z_1 - z_0) \times e^{i\frac{2\pi}{3}}$

$\Rightarrow \quad z_2 = i + (-2i) \times \left(-\dfrac{1}{2} + \dfrac{i\sqrt{3}}{2}\right)$

$\Rightarrow \quad z_2 = i + i(1 - i\sqrt{3}) = 2i + \sqrt{3} = (\sqrt{3}, 2)$

Similarly, we can easily prove that $C = (-\sqrt{3}, 2)$.

122. Let $OA = 3$, so that the complex number A is $3e^{i\frac{\pi}{4}}$.



Let the point P be the complex number z.
Then by the rotation theorem, we have

$$\left(\frac{z-3e^{i\pi/4}}{0-3e^{i\pi/4}}\right)=\frac{4}{3}e^{-i\pi/2}=-\frac{4i}{3}$$

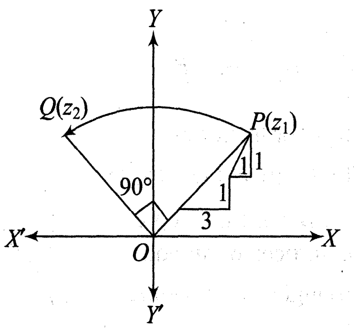
$\Rightarrow \quad 3(z-3e^{i\pi/4})=-4i(-3e^{i\pi/4})$
$\qquad\qquad = 12ie^{i\pi/4}$

$\Rightarrow \quad 3z-9e^{i\pi/4}=12ie^{i\pi/4}$

$\Rightarrow \quad z-3e^{i\pi/4}=4ie^{i\pi/4}$

$\Rightarrow \quad z=(3+4i)e^{i\pi/4}$

123. Let the point P represents the complex number z_1 and Q be z_2.



Here, $z_1=\left(6+\sqrt{2}\cos\left(\frac{\pi}{4}\right), 5+\sqrt{2}\sin\left(\frac{\pi}{4}\right)\right)$
$\qquad = (7, 6)$

By rotation theorem, about the origin

$$\frac{z_2-0}{z_1-0}=\left|\frac{z_2-0}{z_1-0}\right|e^{i\pi/2}=e^{i\pi/2}=i$$

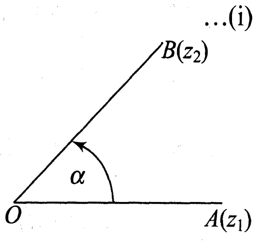
$\Rightarrow \quad z_2=iz_1=i(7+6i)=-6+7i=(-6,7)$

124. Given $z^2+pz+q=0$

Let its roots are z_1, z_2.

$$z_1+z_2=-p,\ z_1z_2=q \qquad \ldots\text{(i)}$$

$$\frac{z_2-0}{z_1-0}=\frac{OA}{OB}e^{i\alpha}=e^{i\alpha}$$



$\Rightarrow \quad \frac{z_2}{z_1}=e^{i\alpha}$

$\Rightarrow \quad z_2=z_1e^{i\alpha} \qquad \ldots\text{(ii)}$

From Eqs (i) and (ii), we get

$$\frac{p^2e^{i\alpha}}{(1+e^{i\alpha})^2}=q$$

$$\Rightarrow \quad \frac{p^2e^{i\alpha}}{[(1+\cos\alpha)+i\sin\alpha]^2}=q$$

$$\Rightarrow \quad \frac{p^2e^{i\alpha}}{\left[2\cos\left(\frac{\alpha}{2}\right)\left(\cos\left(\frac{\alpha}{2}\right)+i\sin\left(\frac{\alpha}{2}\right)\right)\right]^2}=q$$

$$\Rightarrow \quad \frac{p^2e^{i\alpha}}{\left(2\cos\left(\frac{\alpha}{2}\right)\left(e^{i\frac{\alpha}{2}}\right)\right)^2}=q$$

$$\Rightarrow \quad \frac{p^2e^{i\alpha}}{\left[4\cos^2\left(\frac{\alpha}{2}\right)(e^{i\alpha})\right]}=q$$

$$\Rightarrow \quad p^2=4q\cos^2\left(\frac{\alpha}{2}\right)$$

Hence, the result.

125. As we know that, if $\operatorname{Arg}\left(\frac{z_2-z}{z_1-z}\right)=\alpha$, where $\alpha = 0, \pi$, the locus of z is a circle.

Hence, the locus of z in $\operatorname{Arg}\left(\frac{z-1}{z+1}\right)=\frac{\pi}{4}$ is a circle.

126. As we know that the locus of z is an ellipse, if
$|z-z_1|+|z-z_2|=2a$, where $2a>|z_1-z_2|$ and $a\in R^+$.

Hence, the locus of z in $|z-1|+|z+1|\leq 4$ is an ellipse.

127. As we know that, the locus of z is a straight line if
$$|z-z_1|+|z-z_2|=|z_1-z_2|.$$

Hence, the locus of z in $|z-2|+|z+2|\leq 4$ is a straight line.

128. Let $z=x+iy$.

Thus, $x+iy=t+5+i\sqrt{4-t^2}$

Comparing the real and imaginary parts, we get

$$x=t+5,\ y=\sqrt{4-t^2}$$

$\Rightarrow \quad x-5=t,\ y=\sqrt{4-t^2}$

Eliminating t, we get

$$(x-5)^2+y^2=t^2+4-t^2$$

$\Rightarrow \quad (x-5)^2+y^2=4$

Hence, the locus of z is a circle.

129. Let $z=x+iy$.

Now, $\frac{(x+iy)^2}{x+iy-1}=\frac{(x^2-y^2+i2xy)}{(x-1)+iy}$

$$=\frac{(x^2-y^2+i2xy)}{(x-1)+iy}\times\frac{(x-1)-iy}{(x-1)-iy}$$

$$=\frac{(x^2-y^2)(x-1)+4xy^2}{(x-1)^2+y^2}$$

$$+i\frac{2(x-1)xy-y(x^2-y^2)}{(x-1)^2+y^2}$$

Since the given complex number is always real, so its imaginary part is zero.

Thus $\frac{(2(x-1)xy - y(x^2 - y^2))}{(x-1)^2 + y^2} = 0$

$\Rightarrow \quad 2(x-1)xy - y(x^2 - y^2) = 0$

$\Rightarrow \quad 2(x-1)x - (x^2 - y^2) = 0$

$\Rightarrow \quad 2x^2 - 2x - (x^2 - y^2) = 0$

$\Rightarrow \quad x^2 + y^2 - 2x = 0$

$\Rightarrow \quad (x-1)^2 + y^2 = 1$

Thus, the locus of z is a circle.

130. Let $z = x + iy$.

Now,

$$\frac{1}{z} = \frac{1}{x+iy} = \frac{x-iy}{(x+iy)(x-iy)}$$

$$= \frac{x-iy}{x^2+y^2}$$

$$= \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$$

It is given that, $\text{Re}\left(\frac{1}{z}\right) = c$

$\Rightarrow \quad \frac{x}{x^2+y^2} = c$

$\Rightarrow \quad (x^2+y^2) - \frac{x}{c} = 0$

Hence, the locus of z represents an equation of a circle.

131. Let $z = x + iy$.

Then $|(x+iy)^2 - 1| = (x^2 + y^2) + 1$

$\Rightarrow \quad |(x^2 - y^2 - 1) + i(2xy)| = (x^2 + y^2) + 1$

$\Rightarrow \quad \sqrt{(x^2 - y^2 - 1)^2 + 4x^2y^2} = ((x^2 + y^2) + 1)$

$\Rightarrow \quad (x^2 - y^2 - 1)^2 + 4x^2y^2 = ((x^2 + y^2) + 1)^2$

$\Rightarrow \quad (x^2 - y^2)^2 - 2(x^2 - y^2) + 1 + 4x^2y^2$
$= ((x^2 + y^2)^2 + 2(x^2 + y^2) + 1)$

$\Rightarrow \quad ((x^2 + y^2)^2 - 2(x^2 - y^2) + 1)$
$= ((x^2 + y^2)^2 + 2(x^2 + y^2) + 1)$

$\Rightarrow \quad 4x^2 = 0$

$\Rightarrow \quad x = 0$

which represents a straight line.